

Krachttraining in wielrennen

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SPORT VLAANDEREN - SPORTFEDERATIES
UNIVERSITEITEN EN HOGESCHOLEN LO

PACE.
Coaching & Education

Inhoud

1. Rationale
2. Planning & Periodisatie
3. Oefening selectie
4. Testing
5. Ontwikkeling

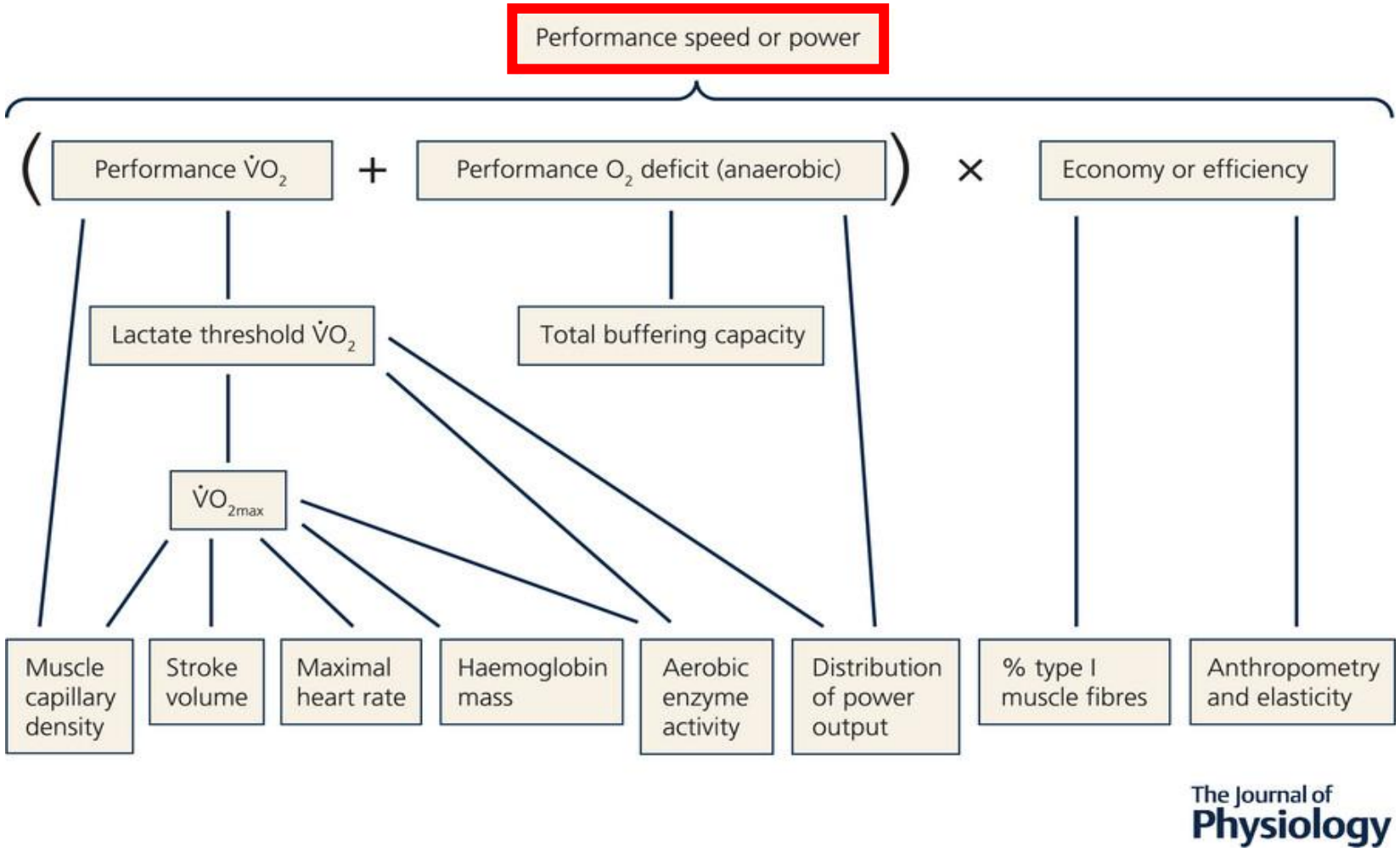
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Table 6. Believed positive effects of strength training, and reasons for why the cyclists perform strength training.

Rationale for performing strength training	Male n = 88	Female n = 59		Total n = 147	Sex differences
	n (%)	*n (%)*	*n (%)*	*V*	*p-value*
To improve cycling performance	76 (86.4)[a]	56 (94.9)[a]	132 (89.8)[a]	.138	n.s.
To reduce injury risk	66 (75)[a]	46 (78)[a]	112 (76.2)[a]	.034	n.s.
For overall health	33 (37.5)[b,c]	25 (42.4)[c,d]	58 (39.5)[b]	.049	n.s.
To increase muscle mass	18 (20.5)[b,c,d]	15 (25.4)[d]	33 (22.4)[c]	.058	n.s.
Coach/team recommendation	17 (19.3)[c,d]	13 (22)[d,e]	30 (20.4)[c]	.033	n.s.
Rehabilitation from injury	14 (15.9)[d,e]	18 (30.5)[c,d]	32 (21.8)[c]	.173	<0.05*
To improve bone health	36 (40.9)[b]	34 (57.6)[b,c]	70 (47.6)[b]	.164	<0.05*
Other reasons	3 (3.4)[e]	2 (3.4)[e]	5 (3.4)[d]	.001	n.s.

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Vikestad et al., 2025

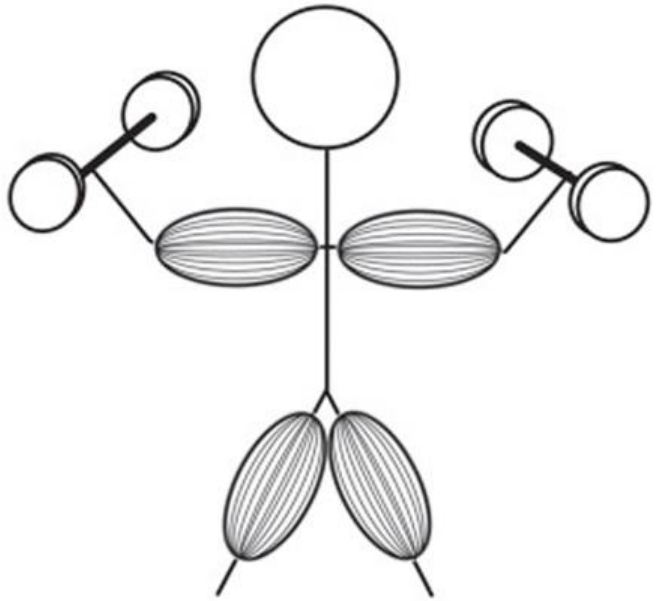
Doel



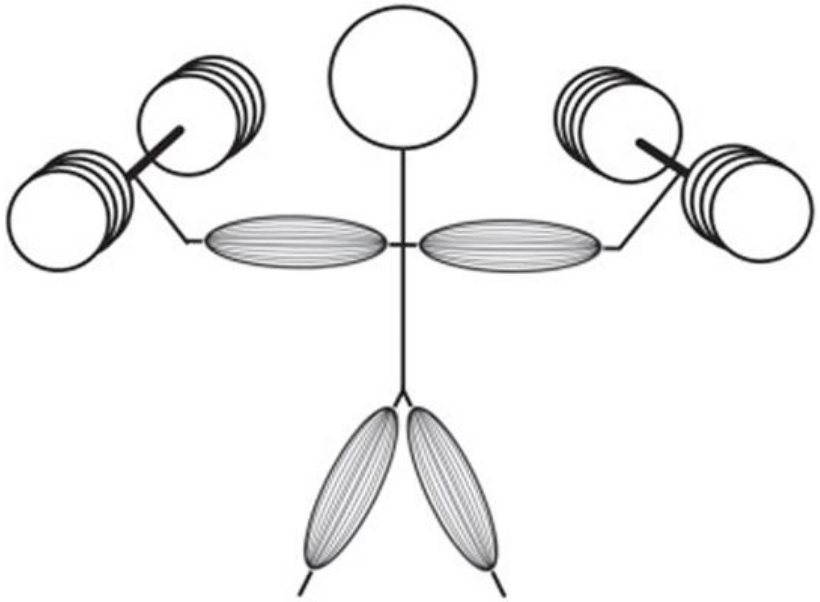
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Joyner & Coyle 2008

Hypertrofie



Verkleinen krachtreserve



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Grenskracht = <u>maximale kracht</u> + krachtreserve

↓ krachreserve = ↑ maximale kracht

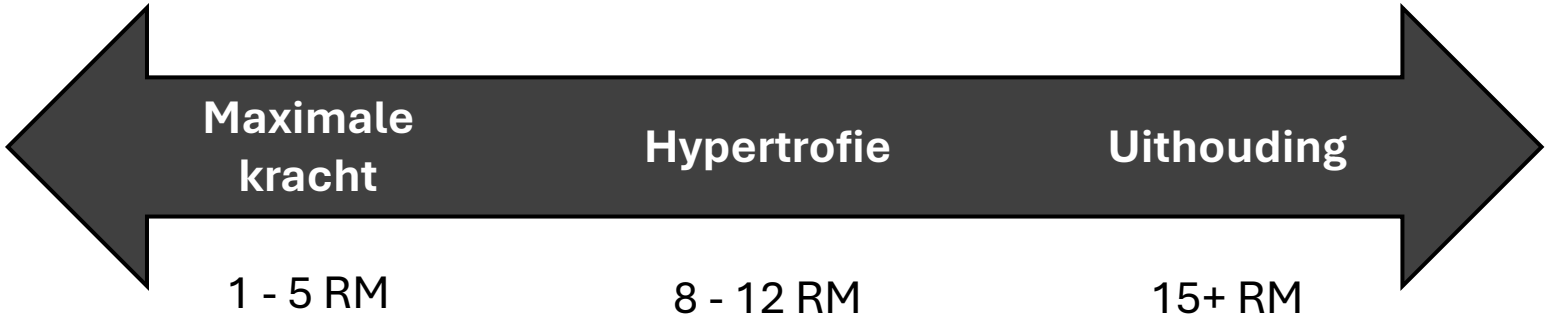


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Krachttraining adaptaties







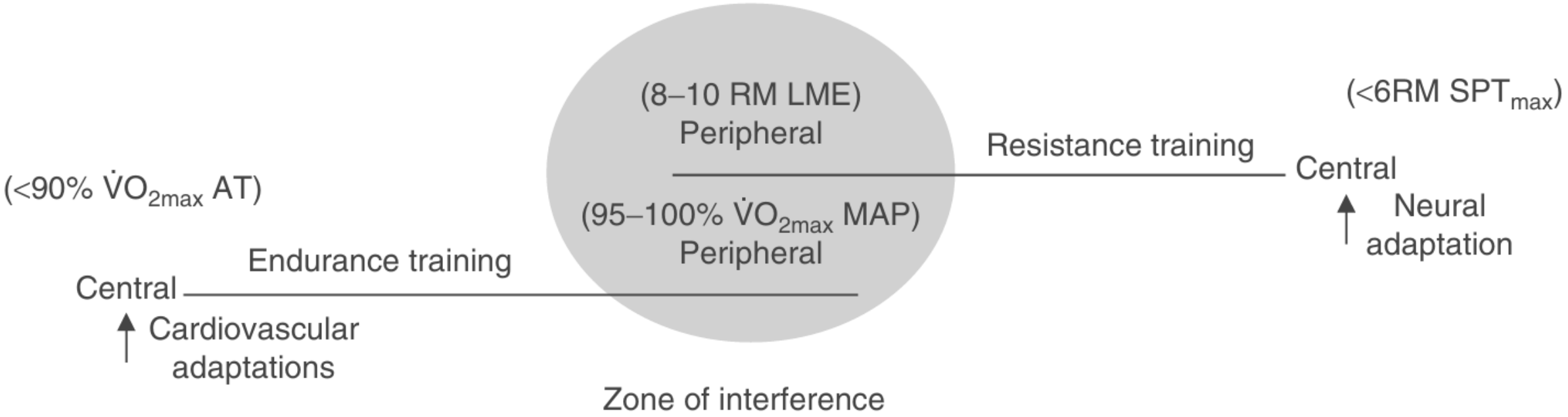
Neuromusculair

- Prikkelgeleiding
- Spiervezelrekrutering
- Intramusculaire coördinatie
- Intermusculaire coördinatie

Morfologisch & metabool

- Hypertrofie
- Hyperplasie
- CP-voorraad
- Glycogeenvoorraad
- Enzymactiviteit

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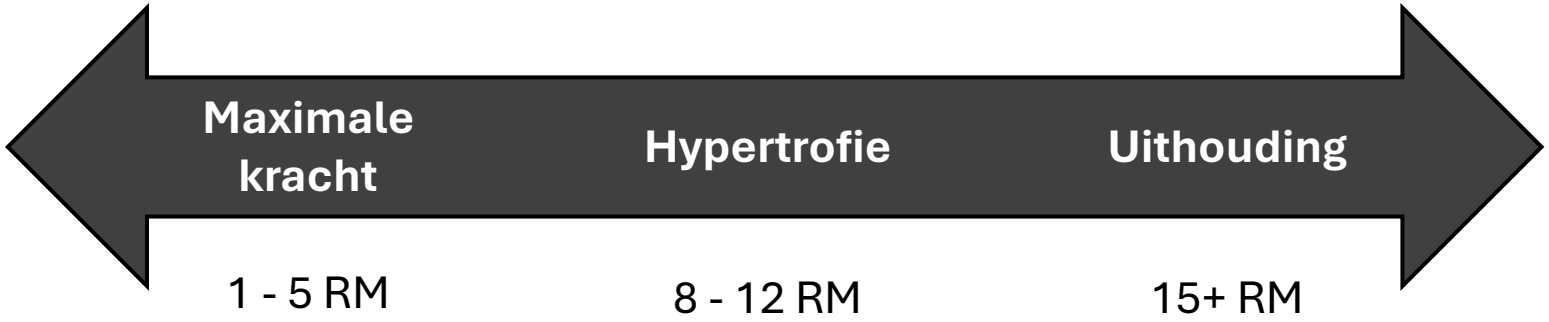


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García-Pallarés & Izquierdo, 2011

Krachttraining adaptaties







Neuromusculair

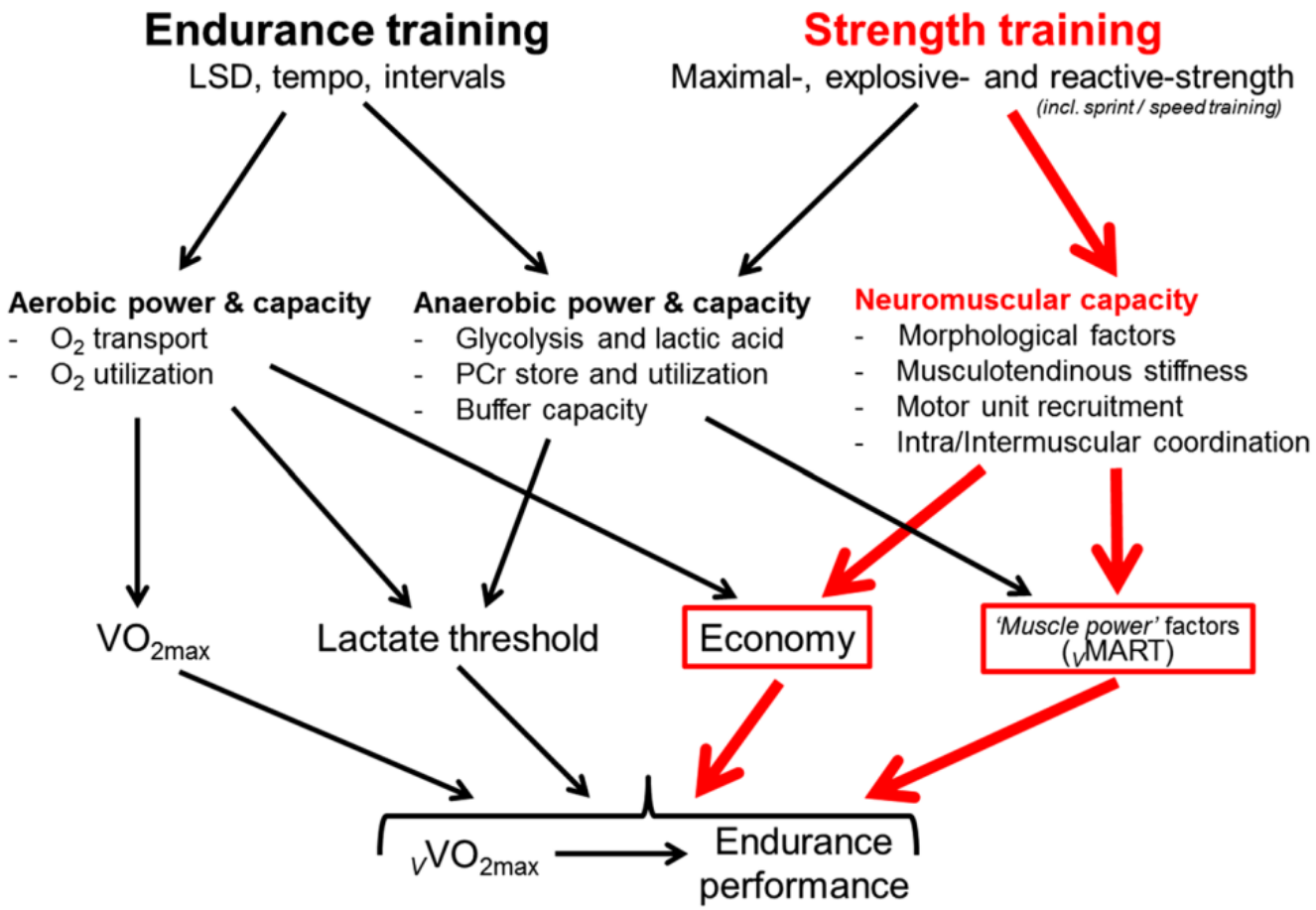
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Morfologisch & metabool

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- Enzymactiviteit

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Bewegingseconomie



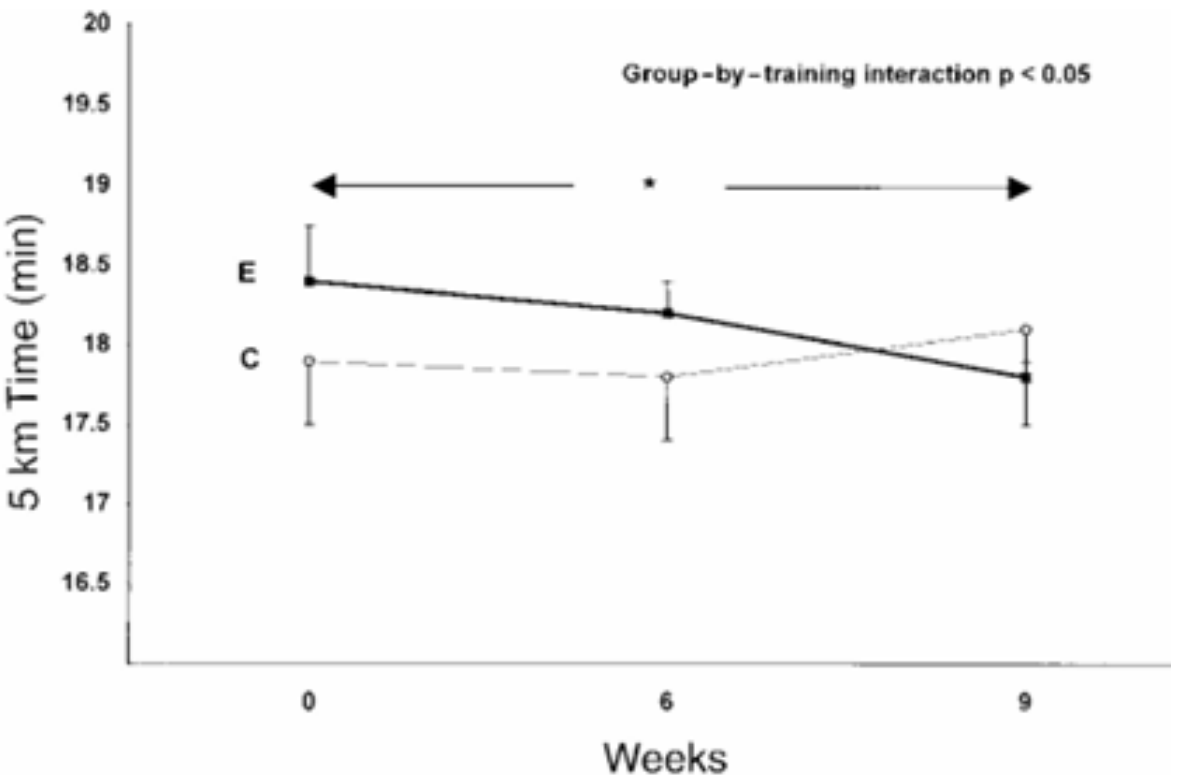
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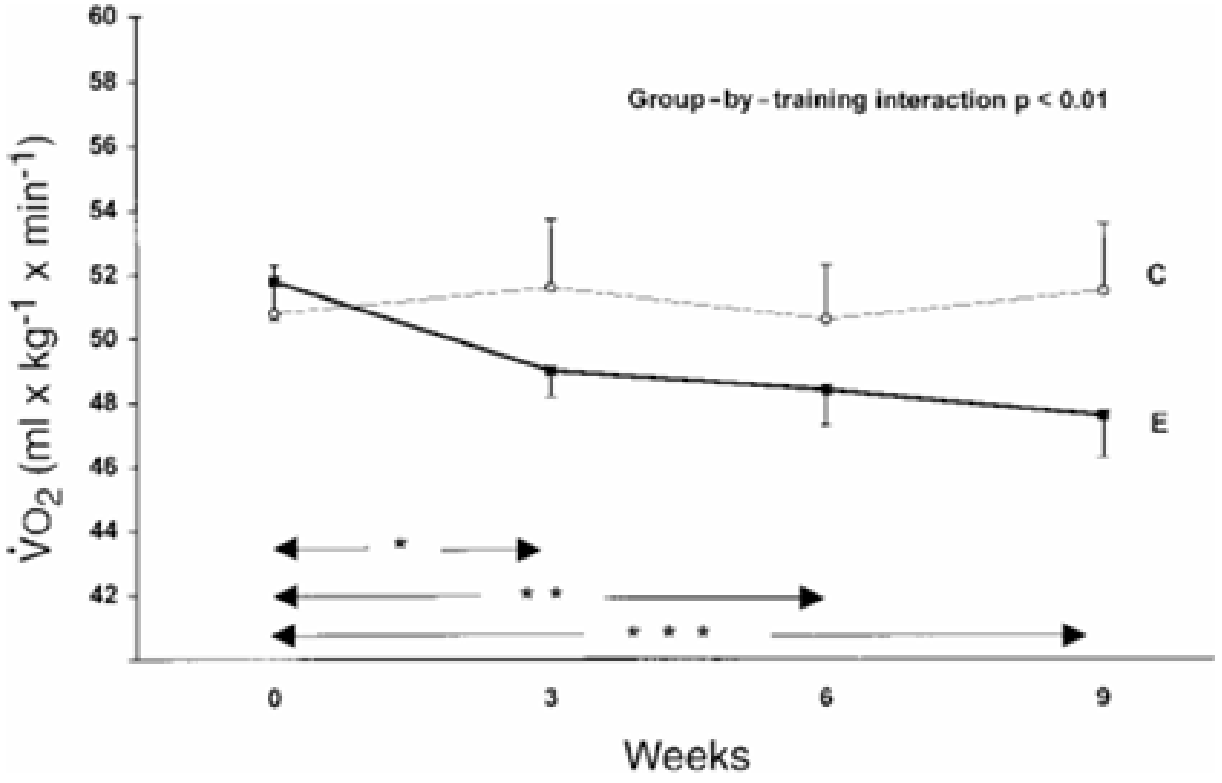
Beattie, Kenny et al. 2014

Trainingsstudie in uithoudingssport

Krachttraining

Controle

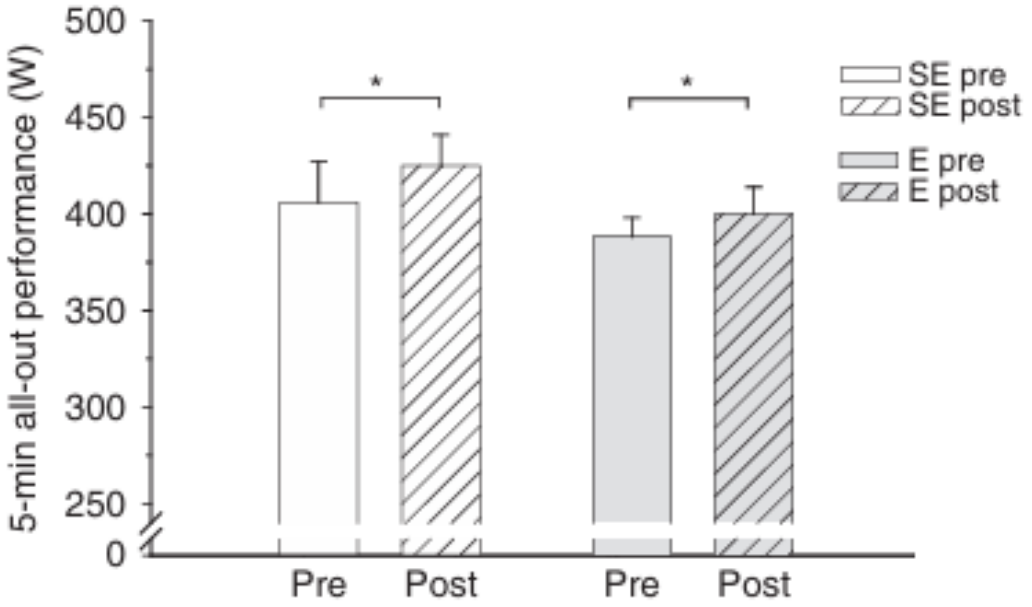


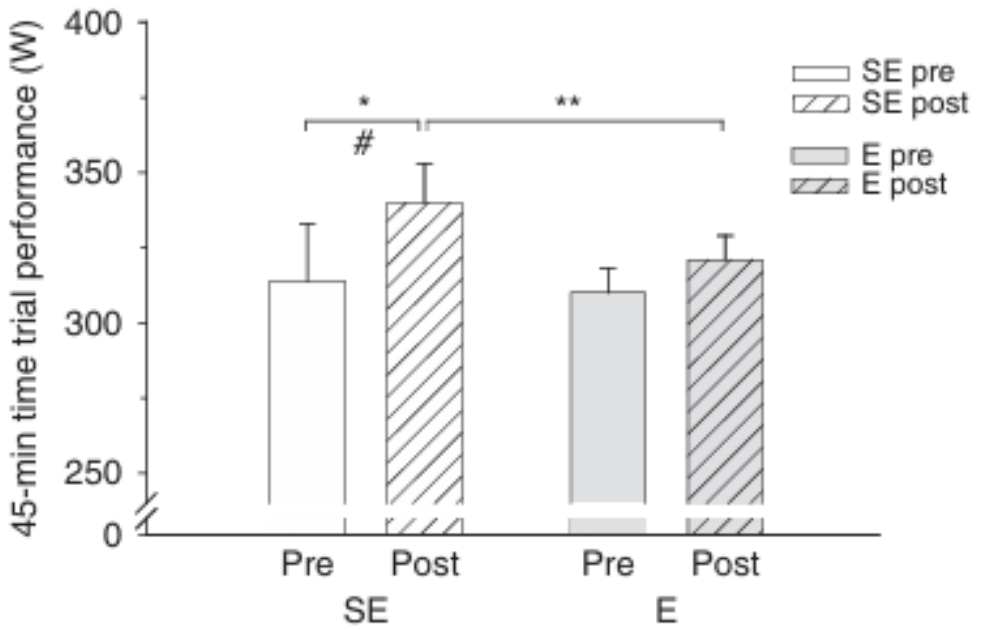


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Paavolainen et al., 1999

Trainingsstudies in wielrennen

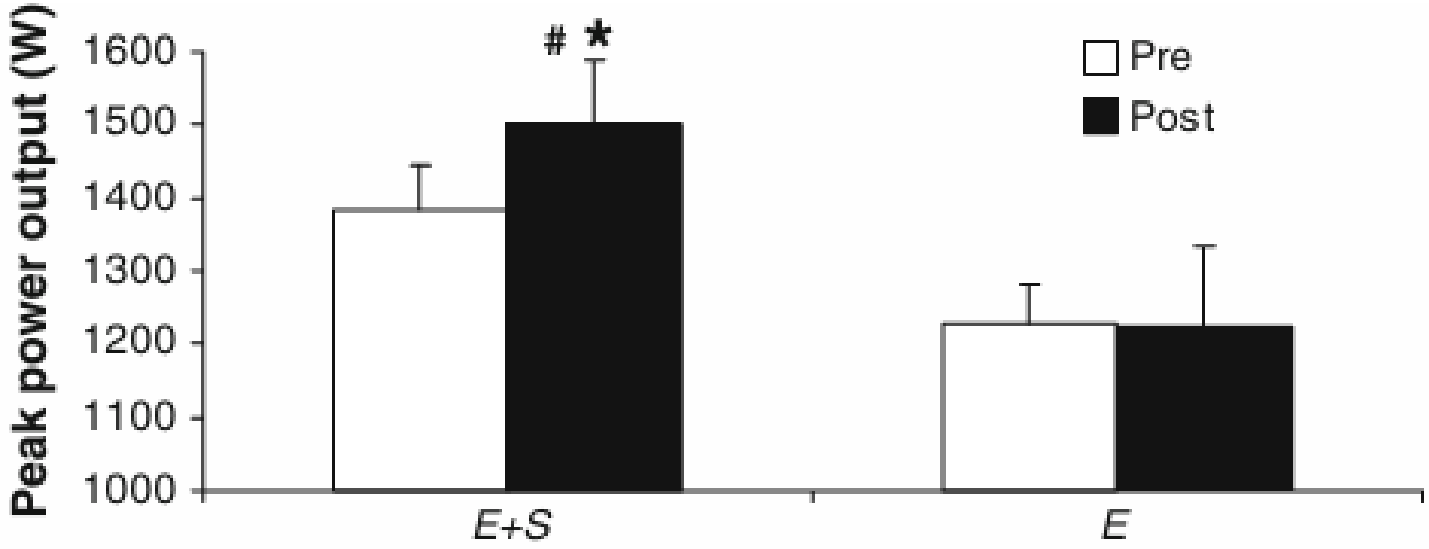




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Aagaard & Andersen, 2010

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Rønnestad et al., 2010 a

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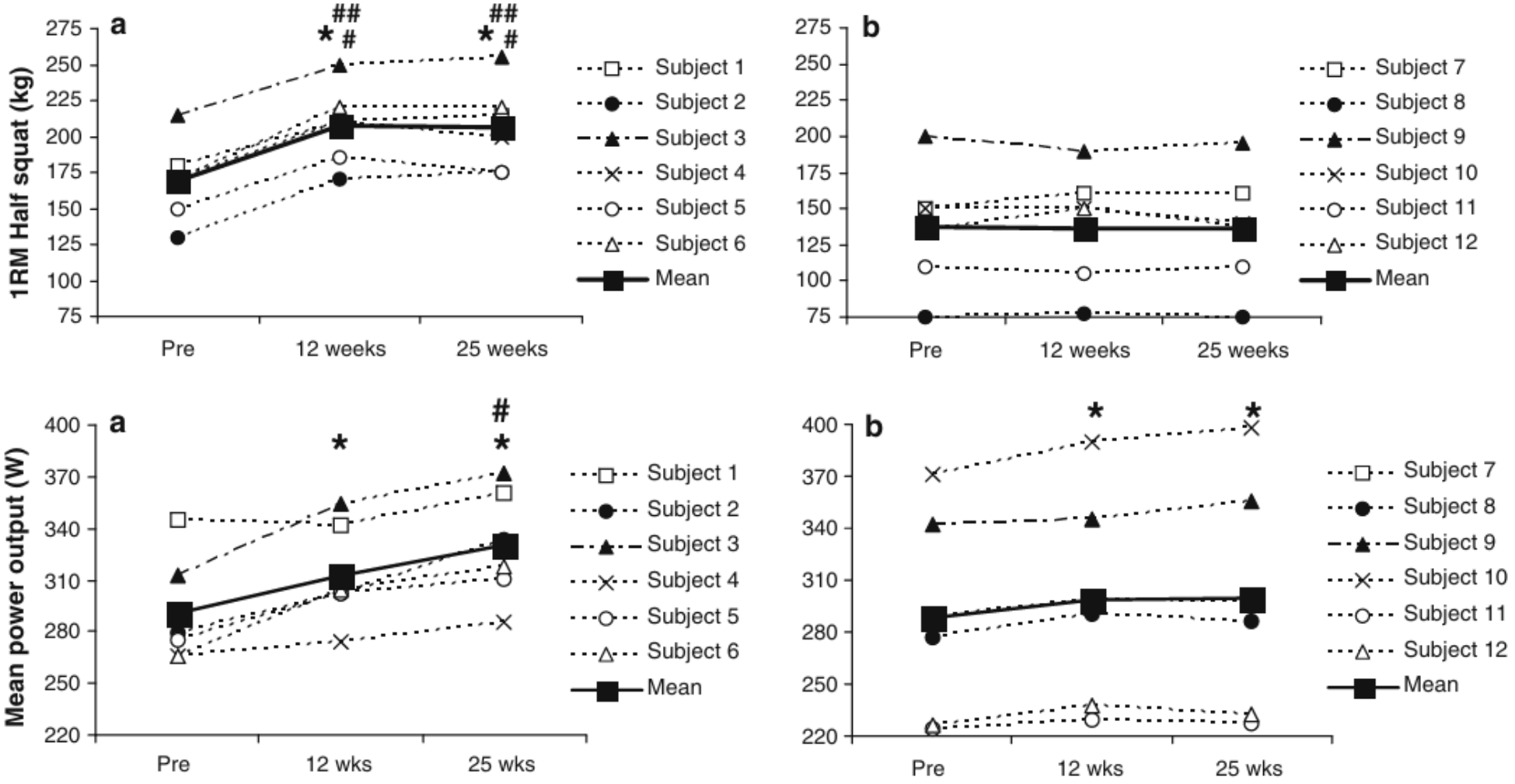
Table 1 Strength training program for the cyclists who performed heavy strength training

	Preparatory period						Competition period
	Week 1–3		Week 4–6		Week 7–12		Week 13–25
	1. Bout	2. Bout	1. Bout	2. Bout	1. Bout	2. Bout	1. Bout
Half squat	3 × 10RM	3 × 6RM	3 × 8RM	3 × 5RM	3 × 6RM	3 × 4RM	2 × 5 reps@80–85% of 1RM
One-legged leg press	3 × 10RM	3 × 6RM	3 × 8RM	3 × 5RM	3 × 6RM	3 × 4RM	2 × 5 reps@80–85% of 1RM
One-legged hip flexion	3 × 10RM	3 × 6RM	3 × 8RM	3 × 5RM	3 × 6RM	3 × 4RM	1 × 6RM
Ankle plantar flexion	3 × 10RM	3 × 6RM	3 × 8RM	3 × 5RM	3 × 6RM	3 × 4RM	1 × 6RM

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Rønnestad et al., 2010 b

Trainingsstudies in wielrennen



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Rønnestad et al., 2010 b

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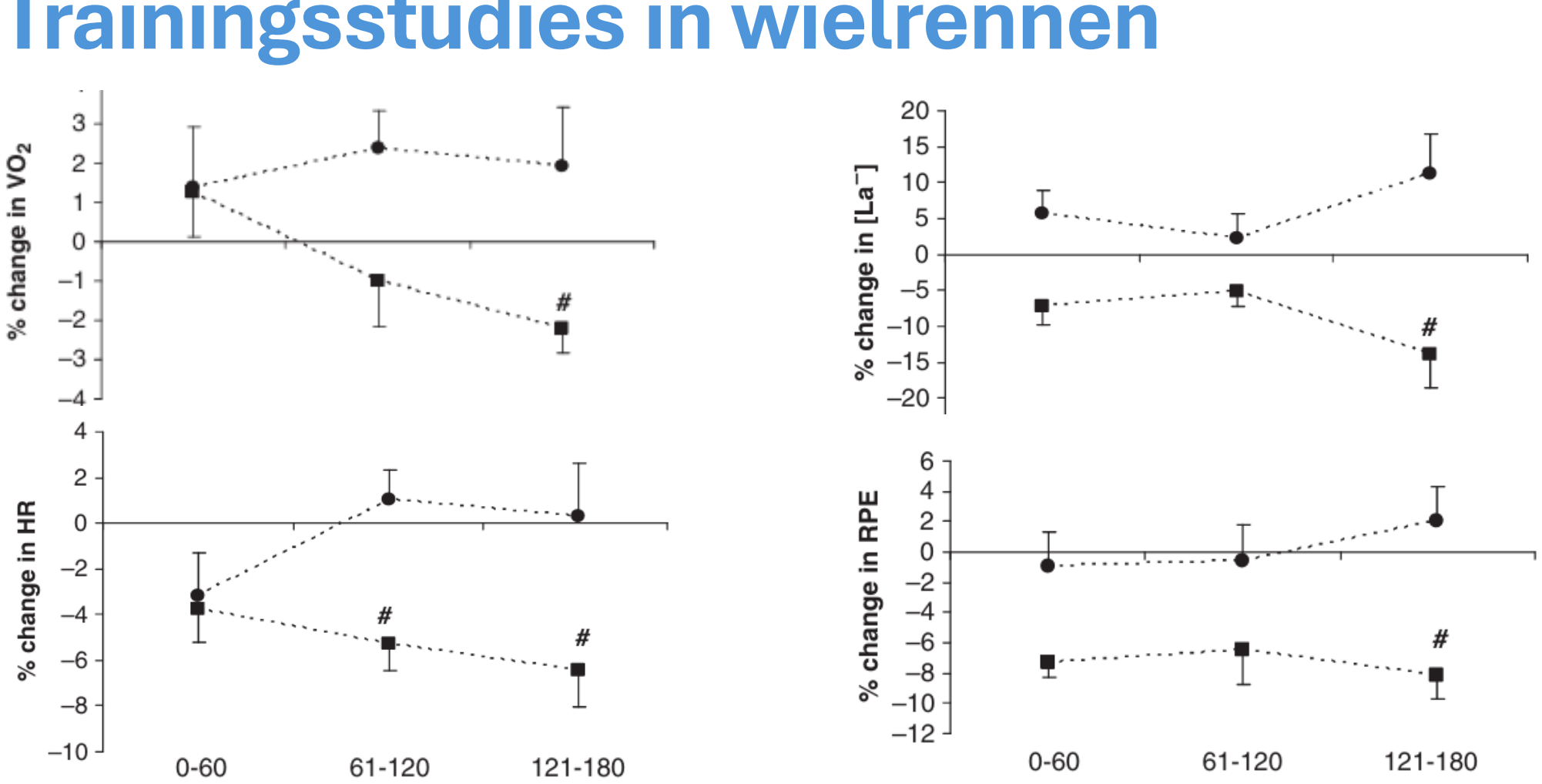
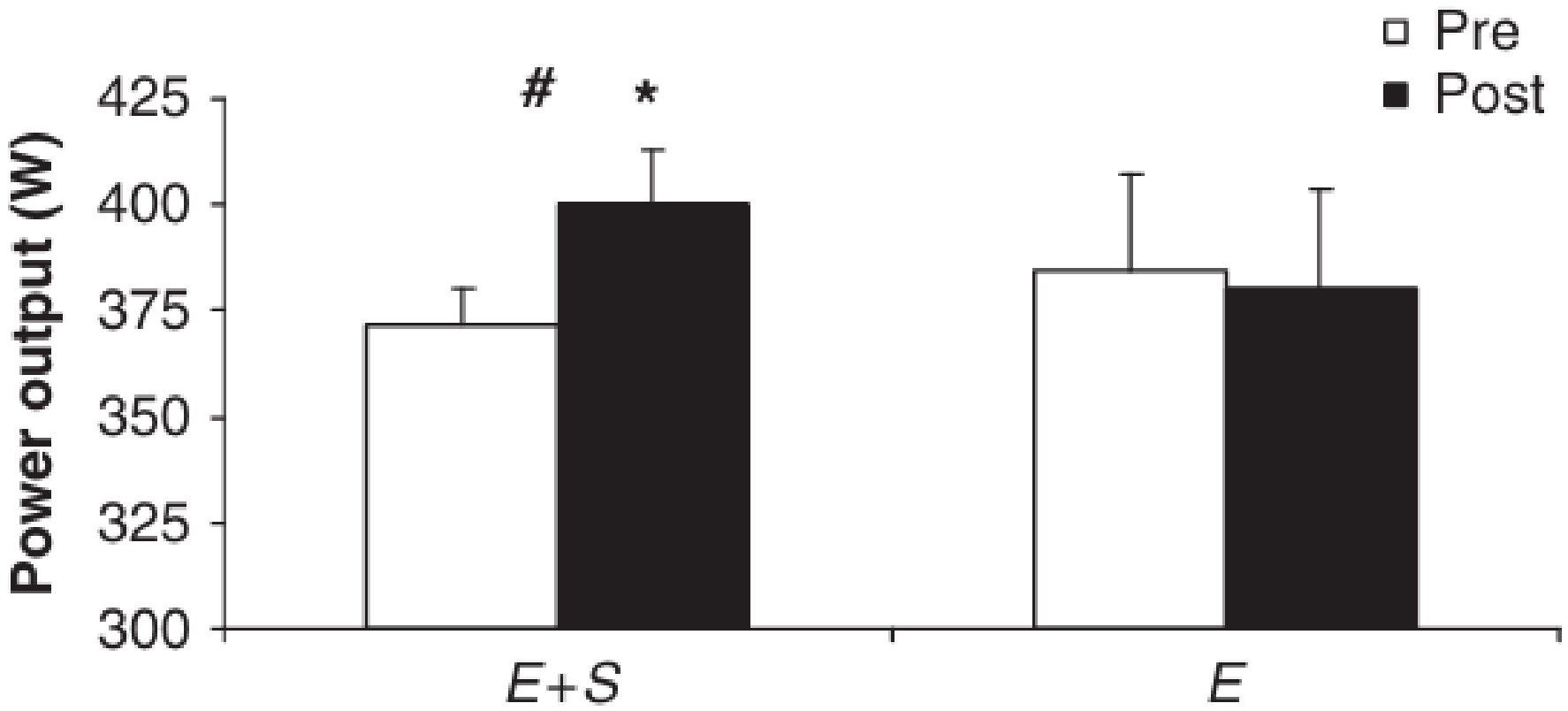


Fig. 4. Relative changes (post- vs pre-intervention) in responses during 180 min of cycling at 44% of baseline W_{max}.

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Rønnestad et al., 2011

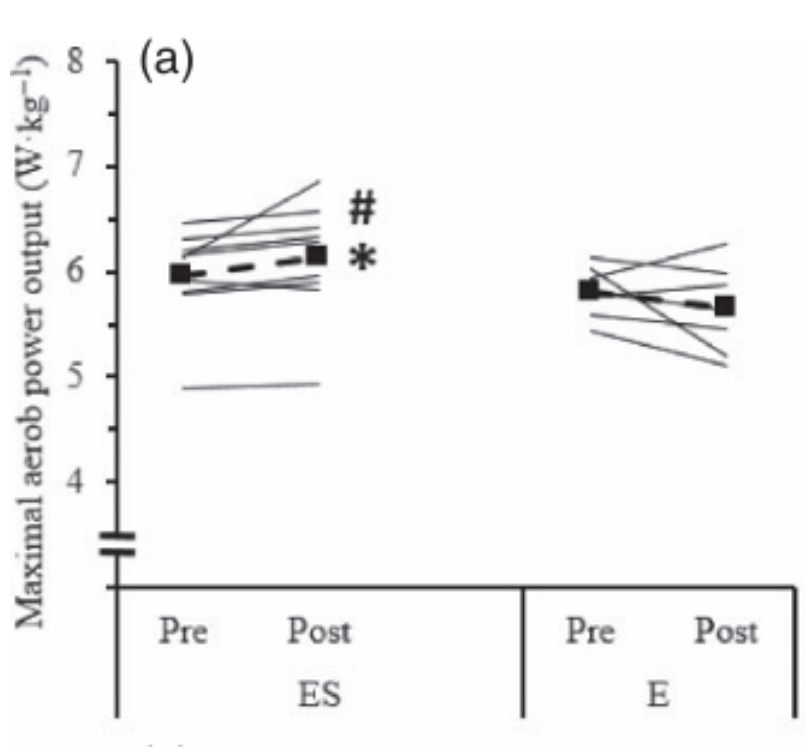
Trainingsstudies in wielrennen

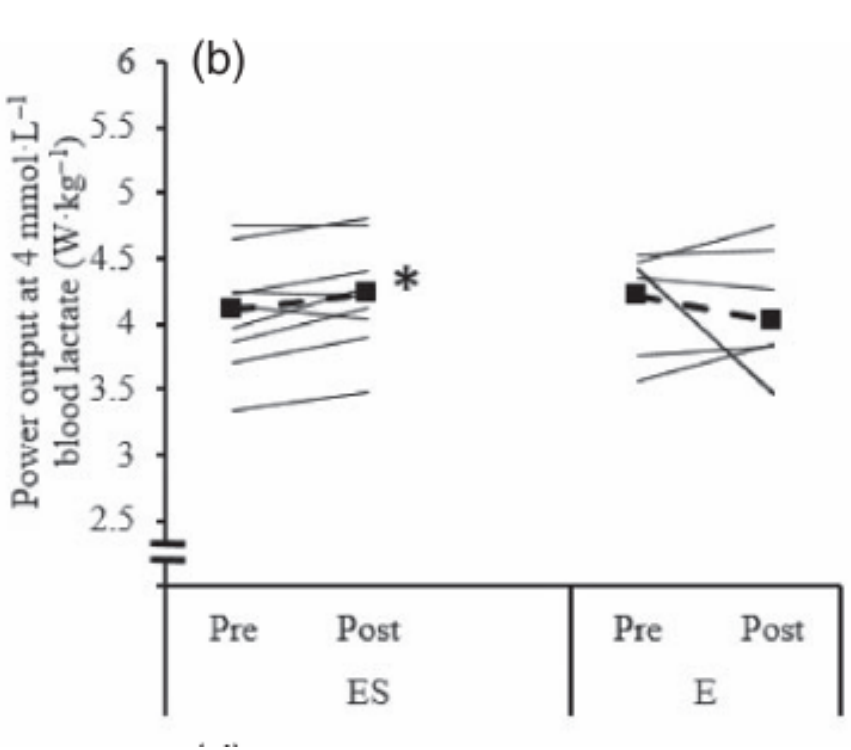


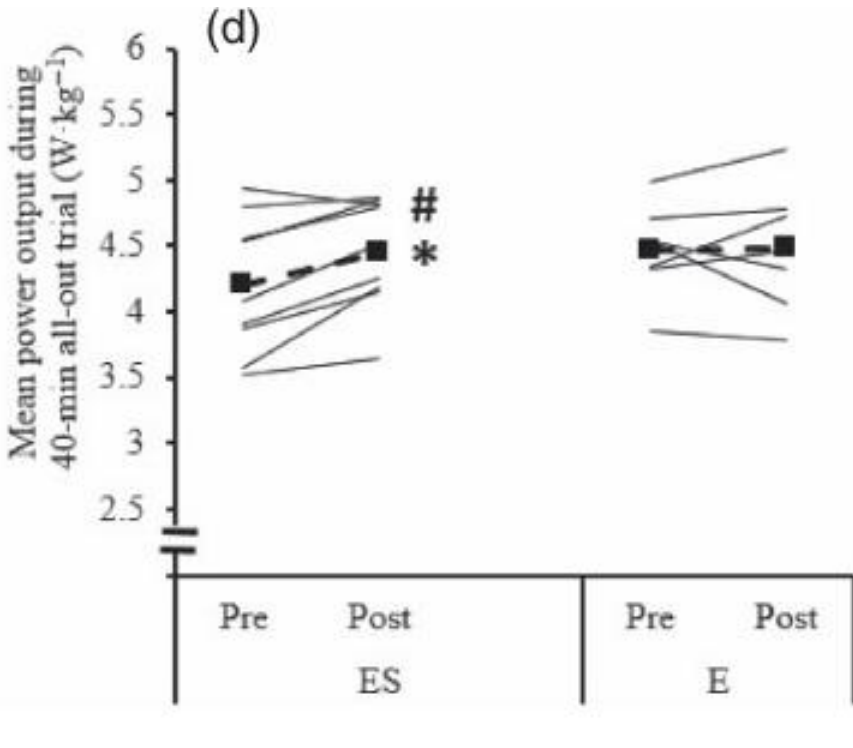
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Rønnestad et al., 2011

Trainingsstudies in wielrennen



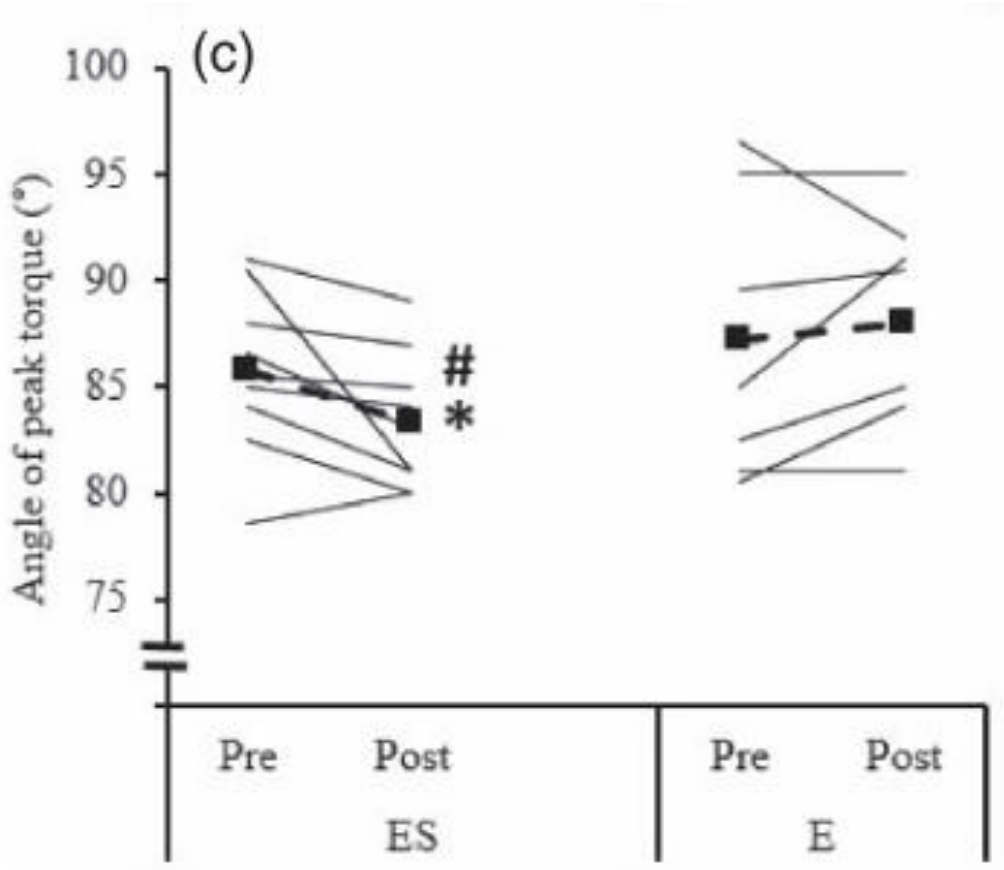




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Rønnestad et al., 2015

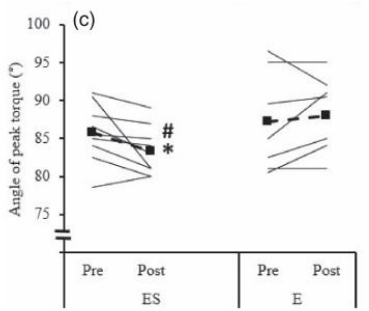
Trainingsstudies in wielrennen

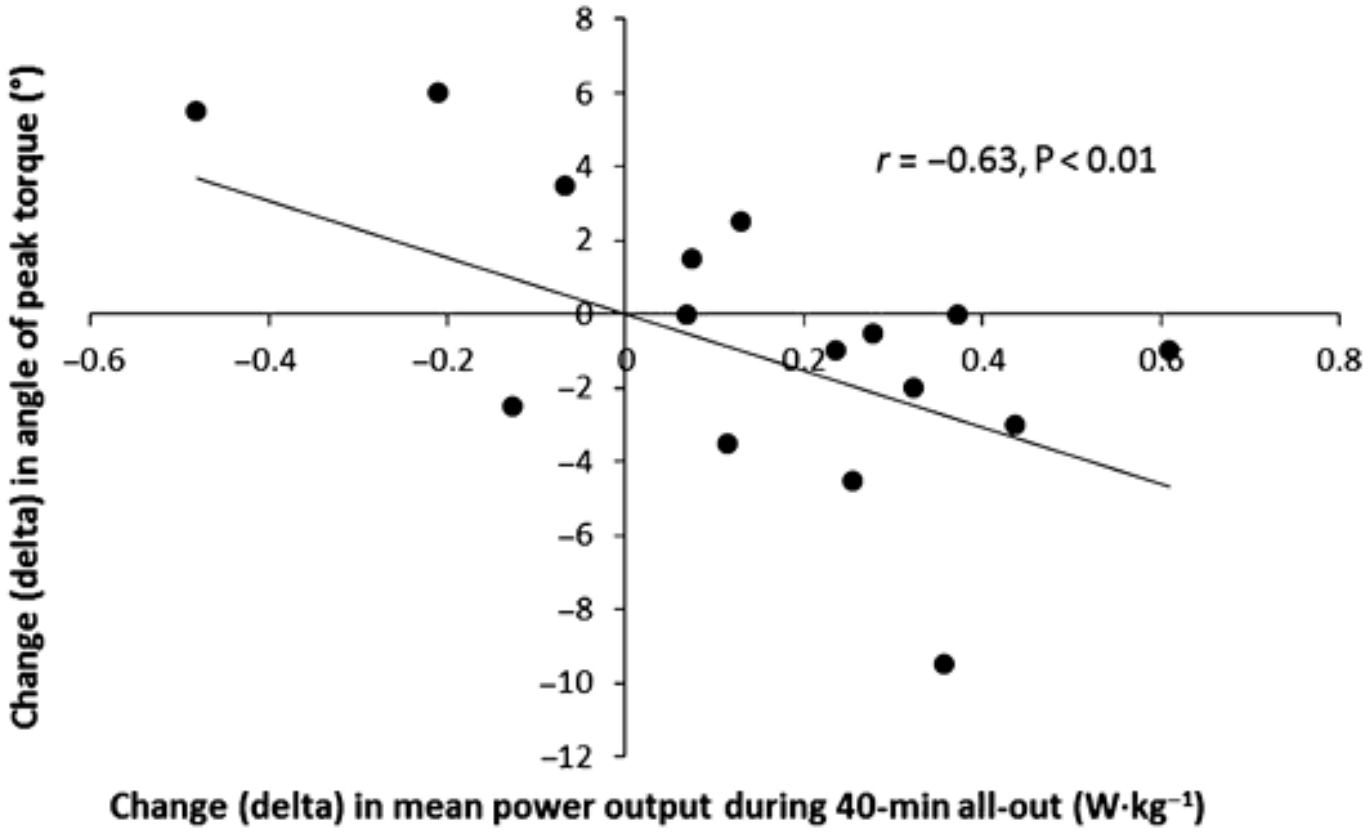


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Rønnestad et al., 2015

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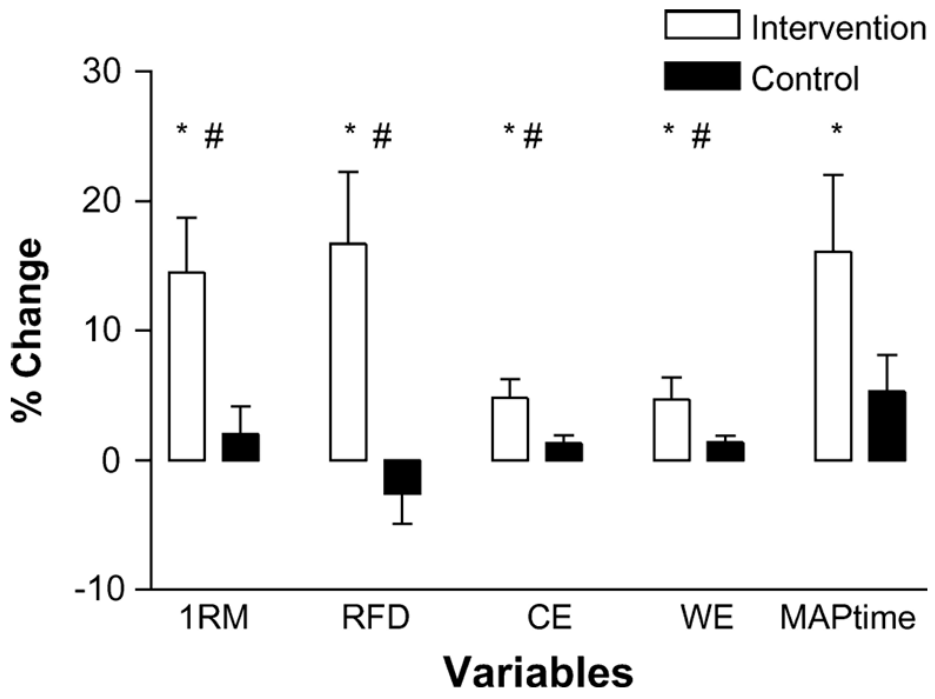


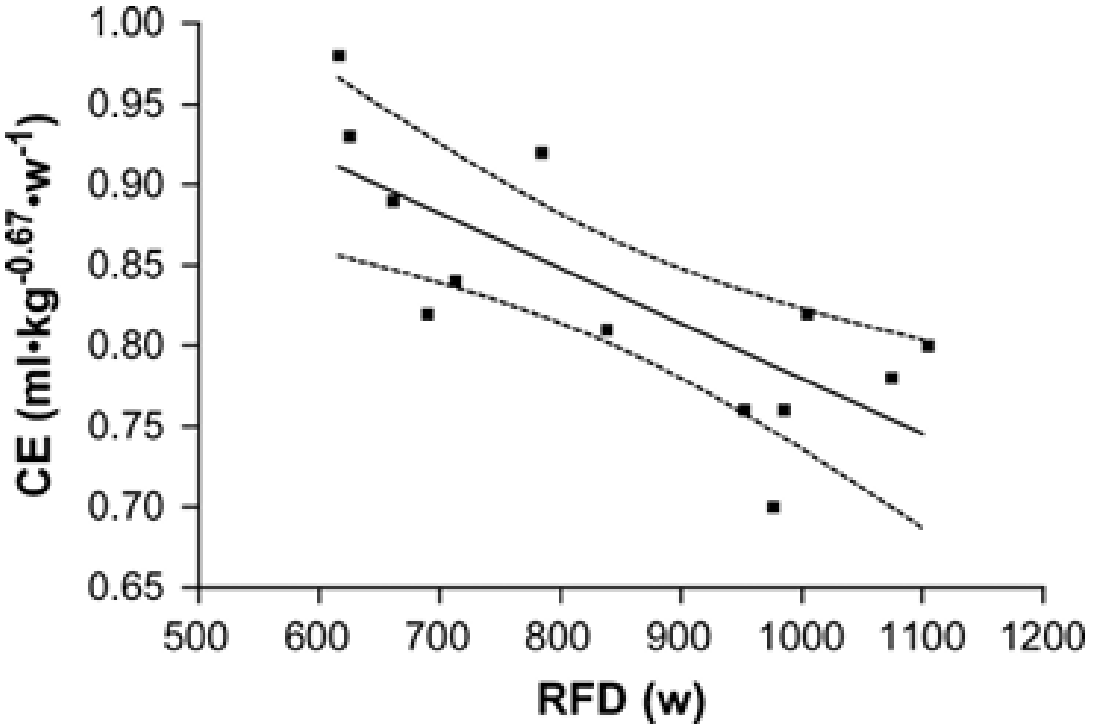


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Rønnestad et al., 2015

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Sunde et al., 2010



HEAVY STRENGTH TRAINING EFFECTS ON PHYSIOLOGICAL DETERMINANTS OF ENDURANCE CYCLIST PERFORMANCE: A SYSTEMATIC REVIEW WITH META-ANALYSIS



Introduction

Endurance cycling performance is determined by maximal oxygen uptake (VO_2max), maximal metabolic steady state (MMSS), non-oxidative energy contribution (i.e., anaerobic capacity and anaerobic power), cycling efficiency and power related to VO_2max (pVO_2max). Heavy strength training (≥ 80% of one repetition maximum) has been shown to have varied effects on performance and its determinants.

Objective and method

Through a systematic review with meta-analysis, we analyzed the effect of heavy strength training on cycling performance and its physiological determinants.

Participant characteristics

♂ 202 ♀ 60

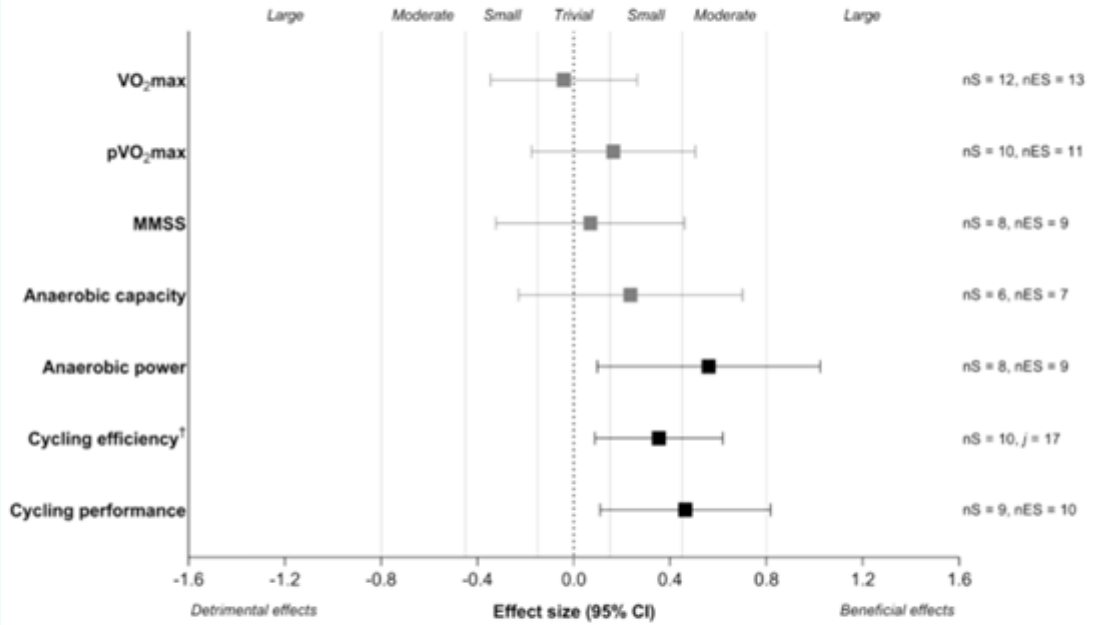
19 to 51 years old

Intervention characteristics

5 a 25 weeks

1 to 3 sessions per week

Results



Summary of the pooled effect size of heavy strength training on physiological determinants (i.e., VO_2max, pVO_2max, MMSS, anaerobic power, anaerobic capacity and cycling efficiency) and cycling performance. Bold boxes and lines represent a significant effect (α = 0.05). CI, confidence interval; j, number of effect sizes within studies; nES, number of effect sizes; nS, number of studies.

†The effect size and 95% CI for cycling efficiency were multiplied by -1 to align the direction of the effect with the other variables.

Conclusion

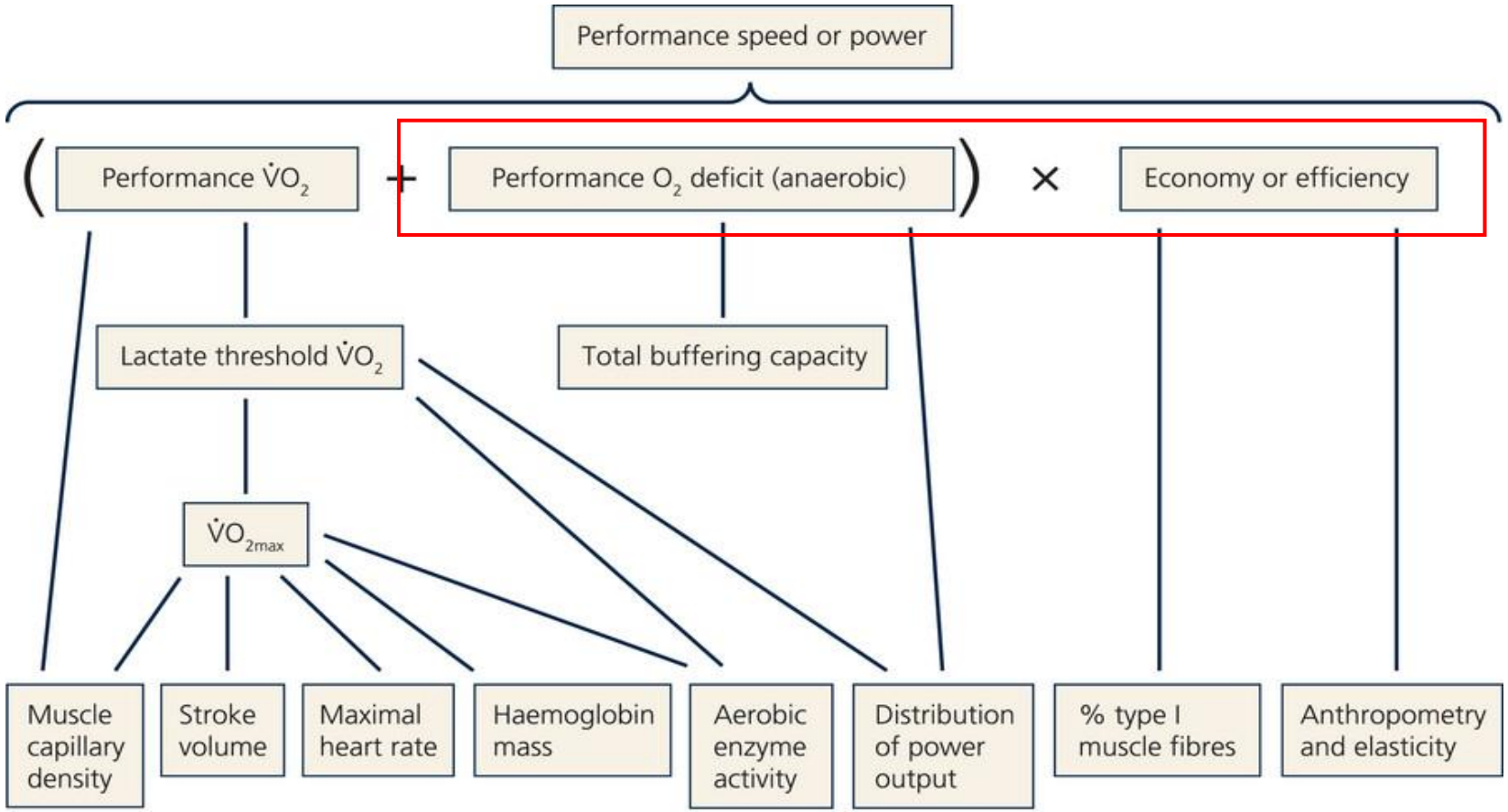
Heavy strength training can improve cycling performance in endurance cyclist. This improvement may be mainly due to an improvement in cycling efficiency and an improvement in anaerobic power. These results occur without changes in VO_2max, pVO_2max, MMSS or anaerobic capacity.

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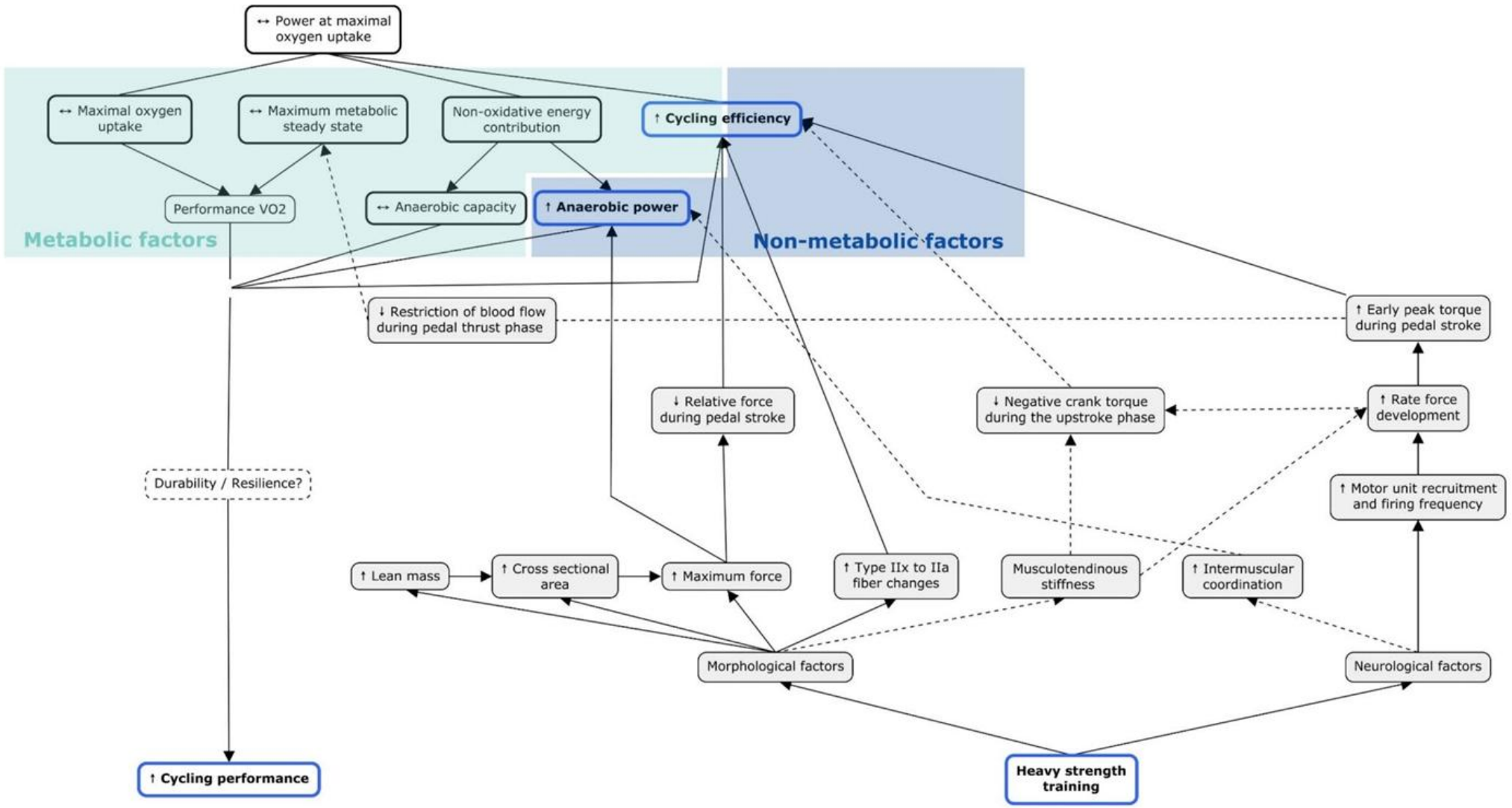
Llanos-Lagos et al., 2026



The Journal of Physiology

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Joyner & Coyle 2008



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Llanos-Lagos et al., 2026

Man-Vrouw

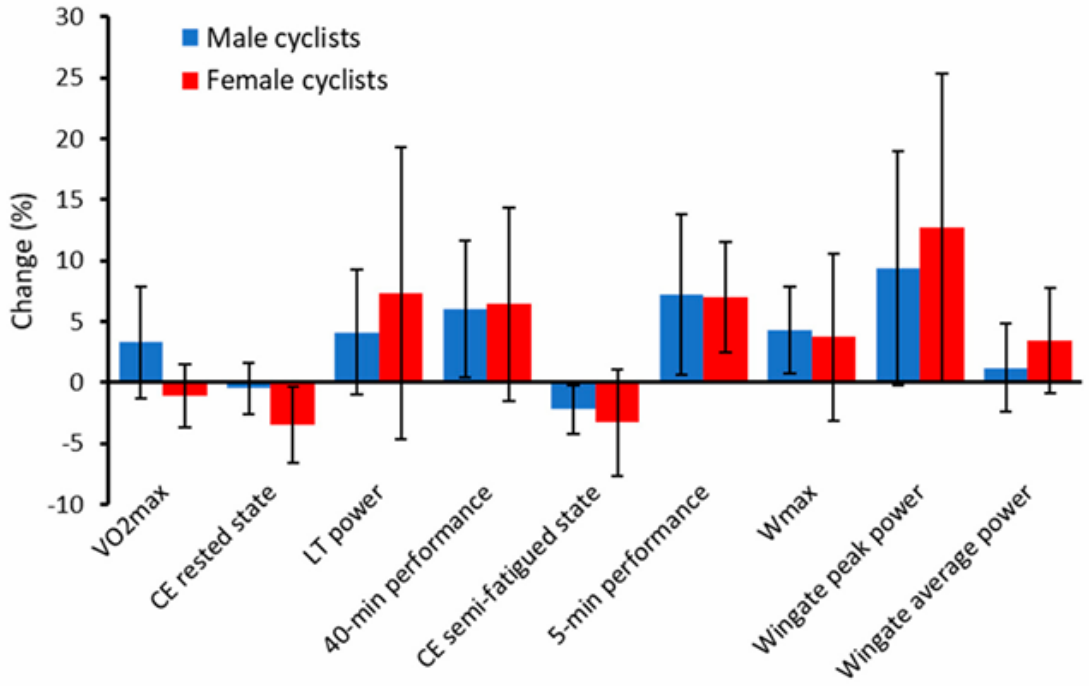
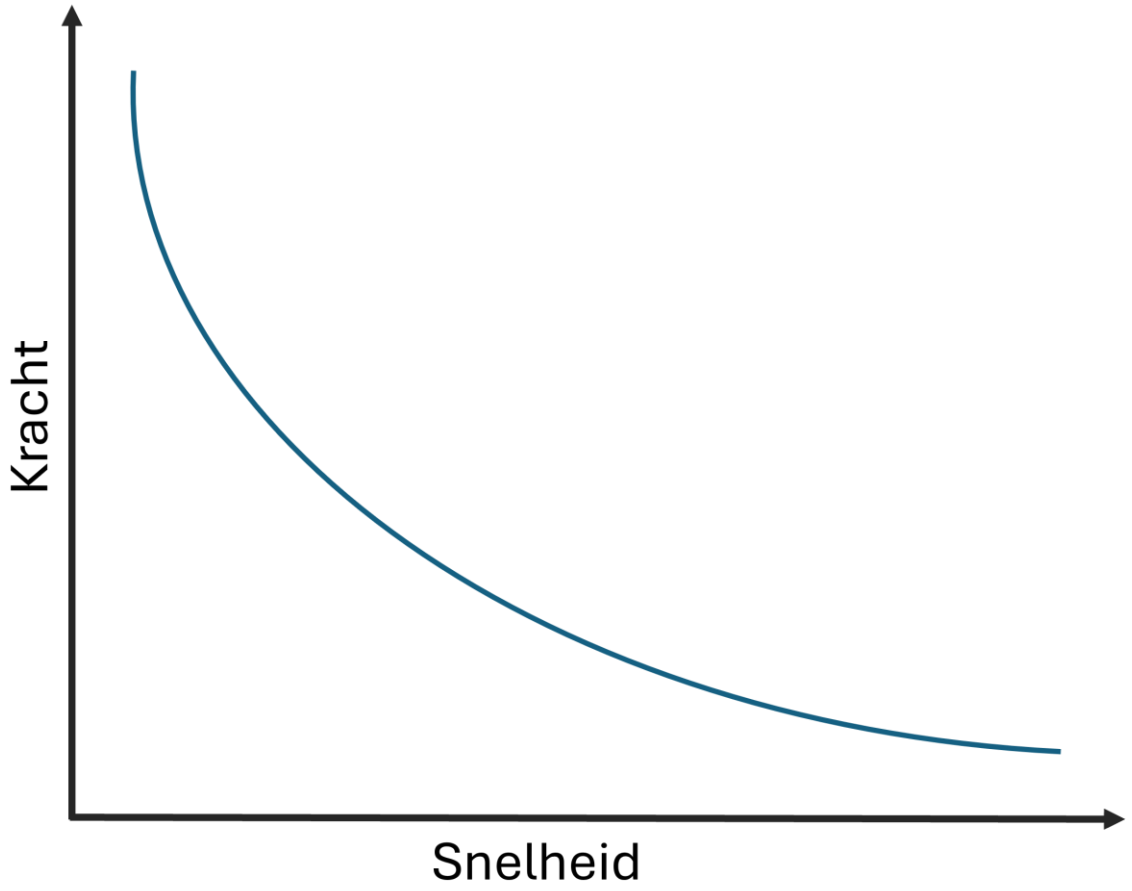


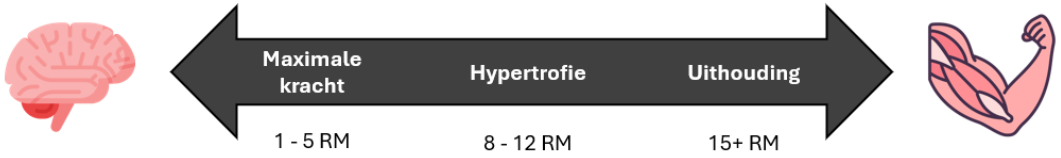
Figure 1. Percent change in physiological measurements and cycling performance after 11–12 weeks of heavy strength training in male and female cyclists. The results from the male cyclists are from [1,16], and the results from the female cyclists are from [2,15]. Values are mean ± SD. RM = repetition maximum, CSA = cross sectional area, VO_{2max} = maximal oxygen consumption, W_{max} = power output at VO_{2max}, CE = cycling economy, LT = lactate threshold. Note: A negative change in cycling economy indicates reduced VO_2 and improved cycling economy. CE semi fatigued state was measured during the last hour of a 3 h submaximal cycling trial, and the 5-min performance was measured directly following the 3 h trial. The increase in VO_{2max} in the male cyclists was not different from control male cyclists.

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(Vikmoen & Rønnestad, 2021)

Zorgen studies voor een bias?





sports
MDPI

Review

Loading Recommendations for Muscle Strength, Hypertrophy, and Local Endurance: A Re-Examination of the Repetition Continuum

Brad J. Schoenfeld [1,*], Jozo Grgic [2], Derrick W. Van Every [1] and Daniel L. Plotkin [1]



Strength

- Strength-related advantages of heavier loads on a dose-dependent basis when testing is carried out on exercises similar to that employed during training
- Minimal differences in strength across loading zones when testing is carried out on a neutral exercise modality

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Schoenfeld et al., 2021

Effect of movement velocity during resistance training on neuromuscular performance

F Pareja-Blanco [1], D Rodríguez-Rosell [1], L Sánchez-Medina [2], E M Gorostiaga [2], J J González-Badillo [1]

Maximal intended velocity training induces greater gains in bench press performance than deliberately slower half-velocity training

Juan José González-Badillo [1], David Rodríguez-Rosell, Luis Sánchez-Medina, Esteban M Gorostiaga, Fernando Pareja-Blanco

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Table 2.2 Neural Adaptations According to Strength Training Zones

Adaptations	INTENSITY ZONES (% OF 1RM)				
	5	4	3	2	1
Intramuscular coordination:	40-60	60-70	70-80	80-90	90-100
• Synchronization	****	***	***	***	****
• Recruitment	**	***	****	****	****
• Rate coding	****	***	***	***	****
Intermuscular coordination	****	***	***	**	*
Disinhibition of inhibitory mechanisms	*	**	***	****	****
Specific hypertrophy	*	**	****	***	**

Adaptation stimulus: **** = very high; *** = high; ** = medium; * = low

All loads are supposed to be moved with the most explosive (and technically correct) concentric action that the load allows.

Used in the Maximum Strength

I	MxS-I	Power
	3-5	
	3-5	
	3-5	1-2
		1-3
		3-5
		3-5
		3-5

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Bompa & Buzzichelli, 2022

Praktische toepassing

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Macro & meso niveau

Preparation				Competitive		Transition
Anatomical adaptation	Hypertrophy if necessary	Maximum strength	Conversion to specific strength (power; power endurance; or muscular endurance short, medium, or long)	Maintenance of maximum strength and specific strength	Cessation of strength training	Compensa-tion training

OCT.	NOV.	DEC.	JAN.	FEB.	MAR.	APR.	MAY	JUNE	JULY	AUG.	SEPT.
Prep. I			Comp. I			T	Prep. II	Comp. II		T	
AA	MxS		Conv. to P	Maint.		AA	MxS	Conv. to P	Maint.	Compens.	

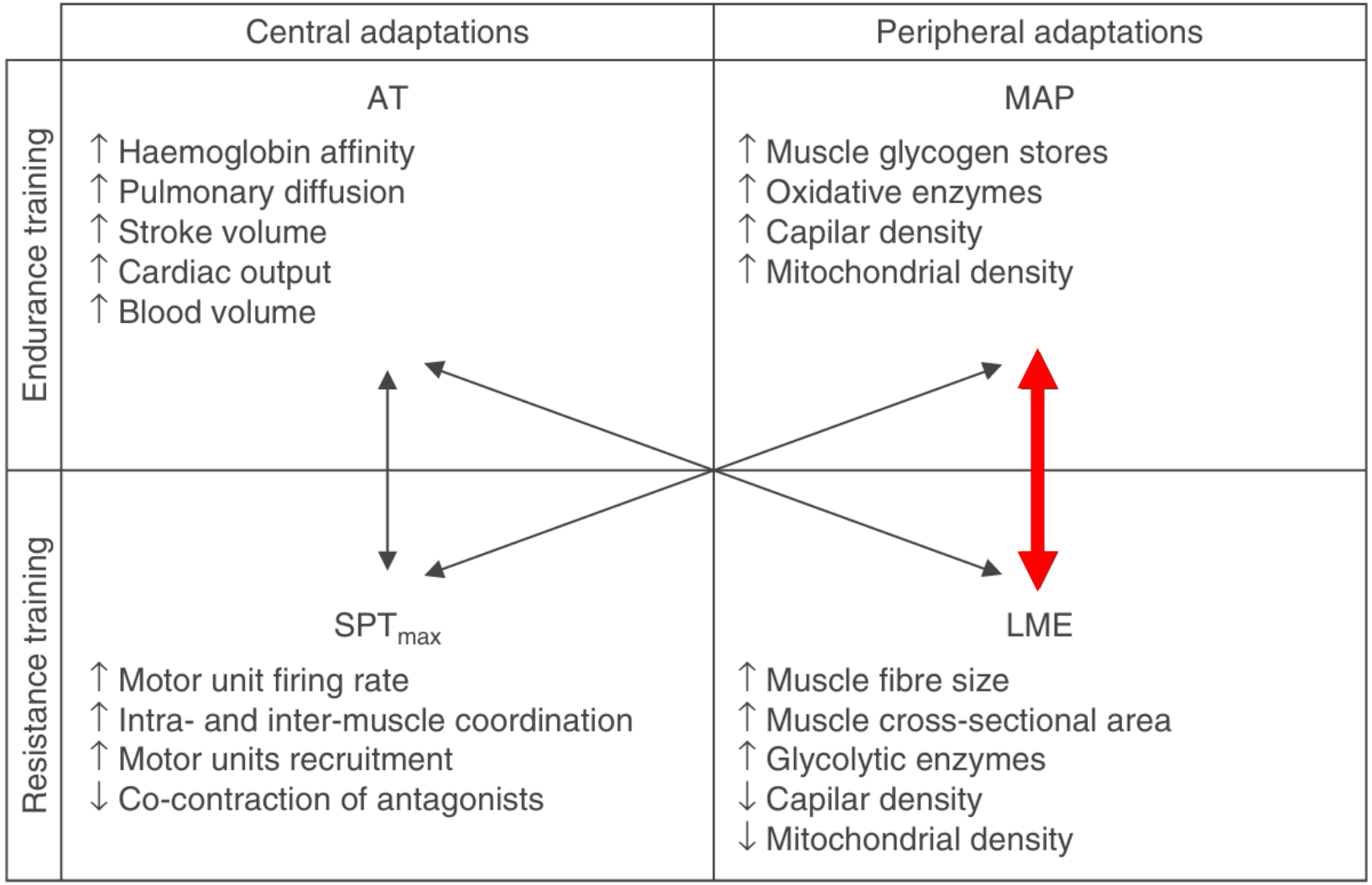
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Bompa & Buzzichelli, 2022

	Adaptatie	Hypertrofie	Maximale kracht	Snelkracht
Intensiteit (%1RM)	40 - 60	60 – 80	≥ 80	30 – 60
Herhalingen	8-15	8-12	≤ 6	4 - 12
Sets	3	3	3	3
Rust tussen sets (min)	3 - 5	1.5 - 2	≥ 3	≥ 3
Uitvoeringssnelheid	2s – 2s	2s – 2s	Maximaal	Maximaal
Volgorde	Verticale seq	Horizontale seq	Verticale seq	Verticale seq

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Bompa & Buzzichelli, 2022

Micro niveau



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García-Pallarés & Izquierdo, 2011

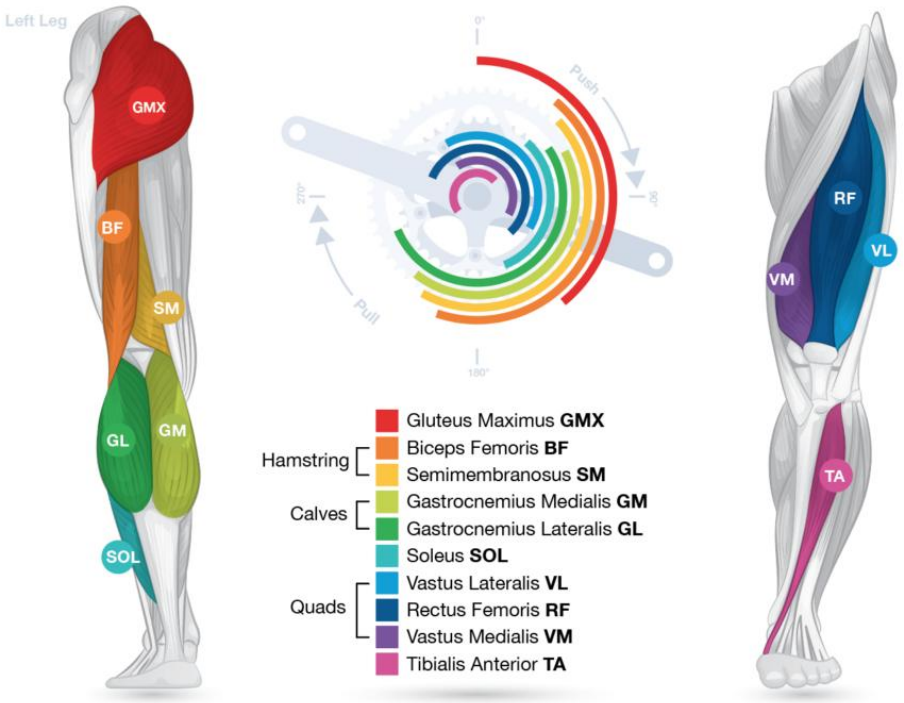
Oefening selectie

Voorbereiding

- Algemeen conditioneren
- ‘Prime movers’+ Accessoire oefeningen

(Pre) Competitie

- Specificiteit
- Quadriceps & Gluteus
- Squat, Hip Thrust, Deadlift, Step Up, Leg Press, …



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1 vs 2 benen





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Total	20h47m	Load	824
Fitness	38	Fatigue	77
Form	-39	Ramp	14.4

Time 2h0m

Time 18h2m
Load 790

Time 45m
Load 34

1h0m	3h30m	1h0m	1h0m	4h20m	5h0m	45m
VM: Kracht	Load 149	Load 25	VM: Kracht	Load 179	Load 212	Load 34
2h2m	Duur	Recovery	2h10m	Duur + 4x15' FatMax	Duur	easy run
Load 87			Load 138			core + mobility
NM: Torque			NM: Duur + sprints			

Total	26h52m	Load	1092
Fitness	45	Fatigue	109
Form	-64	Ramp	20.6

Time 2h0m

Time 24h52m
Load 1092

Zones Polarized 2.40

1h0m	3h3m	6h0m	1h0m	1h0m	6h0m	5h0m
VM: Kracht	Load 136	Load 264	Load 25	VM: Kracht	Load 264	Load 212
1h0m	VO2max efforts #VO2max	Duur	Recovery/Rest	2h49m	Duur	Duur
Load 25				Load 166		
Recovery				VO2max		

Total	20h0m	Load	355
Fitness	32	Fatigue	30
Form	2	Ramp	3.2

Time 19h0m
Dist 400 km
Load 355

Time 1h0m

Load 25	Load 132	Load 132	Load 25	Load 41	5h0m 200 km (A)	5h0m 200 km (A)
1h0m	1h0m	3h0m	1h0m	1h0m	RACE	RACE
Load 25	VM: Kracht	Load 132	Load 25	Load 41	47d	48d
Recovery/Rest	3h0m	Duur	Recovery/Rest	pre race activation		
	Load 132					
	Duur					

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Voorbeelden

B1 Single Leg Hip Thrust
2 SETS
8 @ 25 kg
8 @ 25 kg

B2 Goblet Squat
2 SETS
12 @ 20 kg
12 @ 20 kg

B3 Kettlebell tibialis raises
2 SETS
7 @ 3 kg
7 @ 3 kg

B4 DB Walking Lunge
2 SETS
10 @ 20 kg
10 @ 20 kg

B5 Back Extension Isometric Hold
2 SETS
3 @ 15 kg, 00:00:15
3 @ 15 kg, 00:00:15

B1 Back Squat
3 SETS
4 @ 100 kg
4 @ 100 kg
4 @ 100 kg

B2 Barbell Hip Thrust
3 SETS
4 @ 110 kg
4 @ 110 kg
4 @ 110 kg

B3 Barbell Step Ups
3 SETS
4 @ 55 kg
4 @ 55 kg
4 @ 55 kg

B4 single leg box jump
2 SETS
6
6

B1 Box Squat
3 SETS
5 @ 50 kg
5 @ 50 kg
5 @ 50 kg

B2 Barbell Hip Thrust
3 SETS
5 @ 70 kg
5 @ 70 kg
5 @ 70 kg

B3 Barbell Step Ups
3 SETS
5 @ 30 kg
5 @ 30 kg
5 @ 30 kg

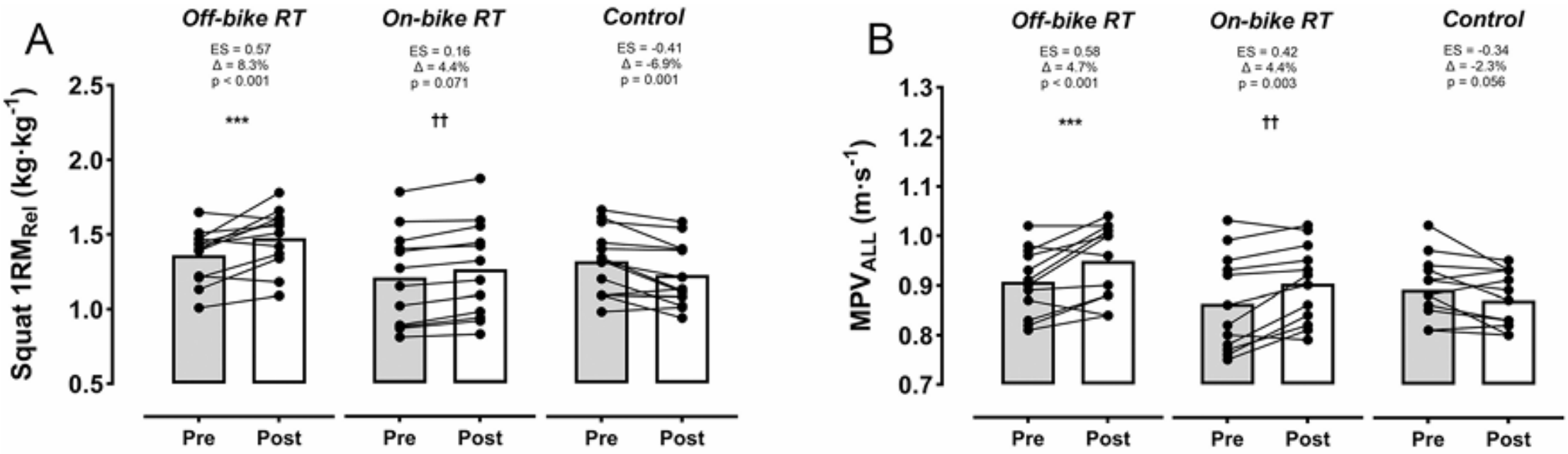
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Sportspecifiek



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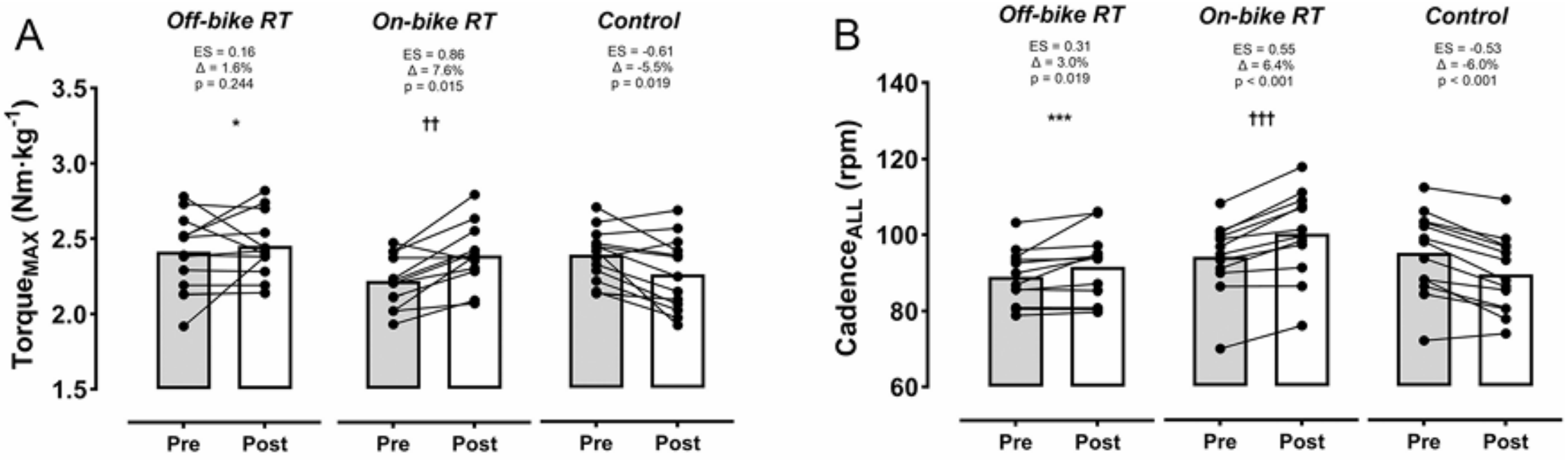
Gym vs bike



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(Barranco-Gil et al., 2025)

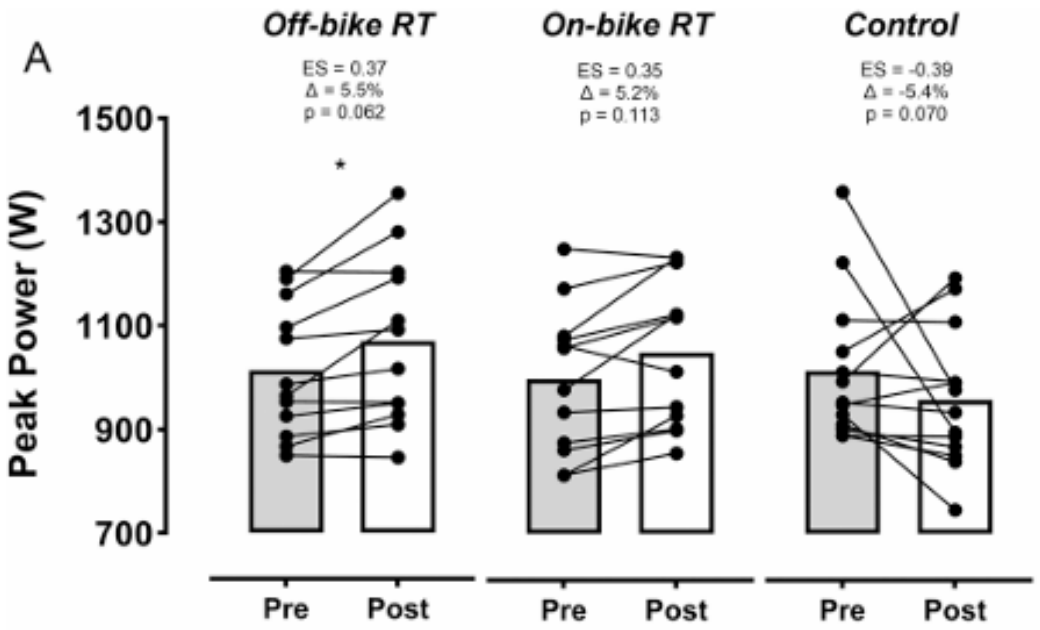
Gym vs bike

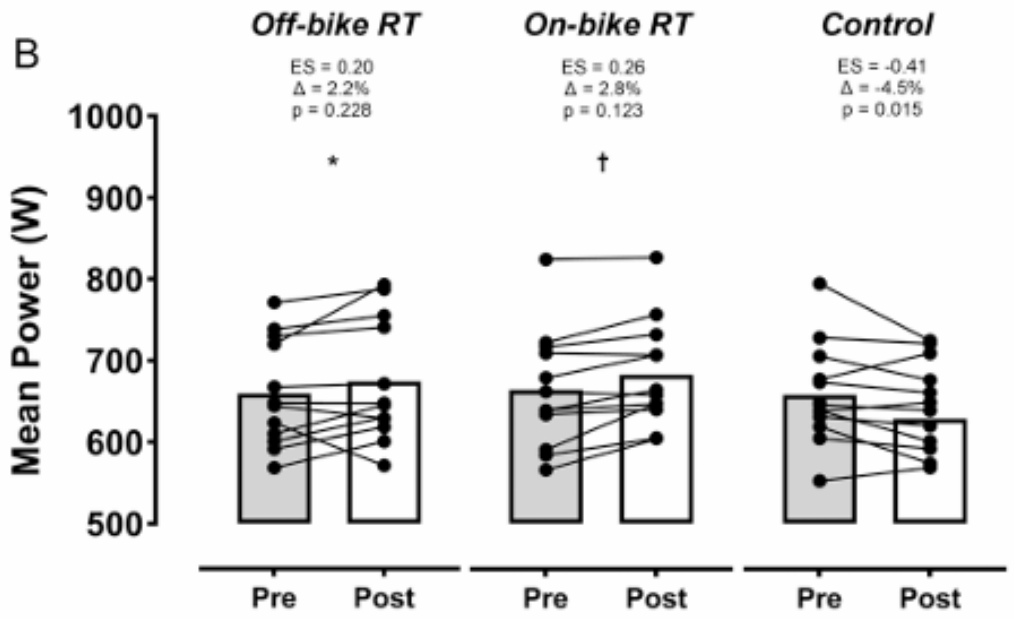


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(Barranco-Gil et al., 2025)

Gym vs bike

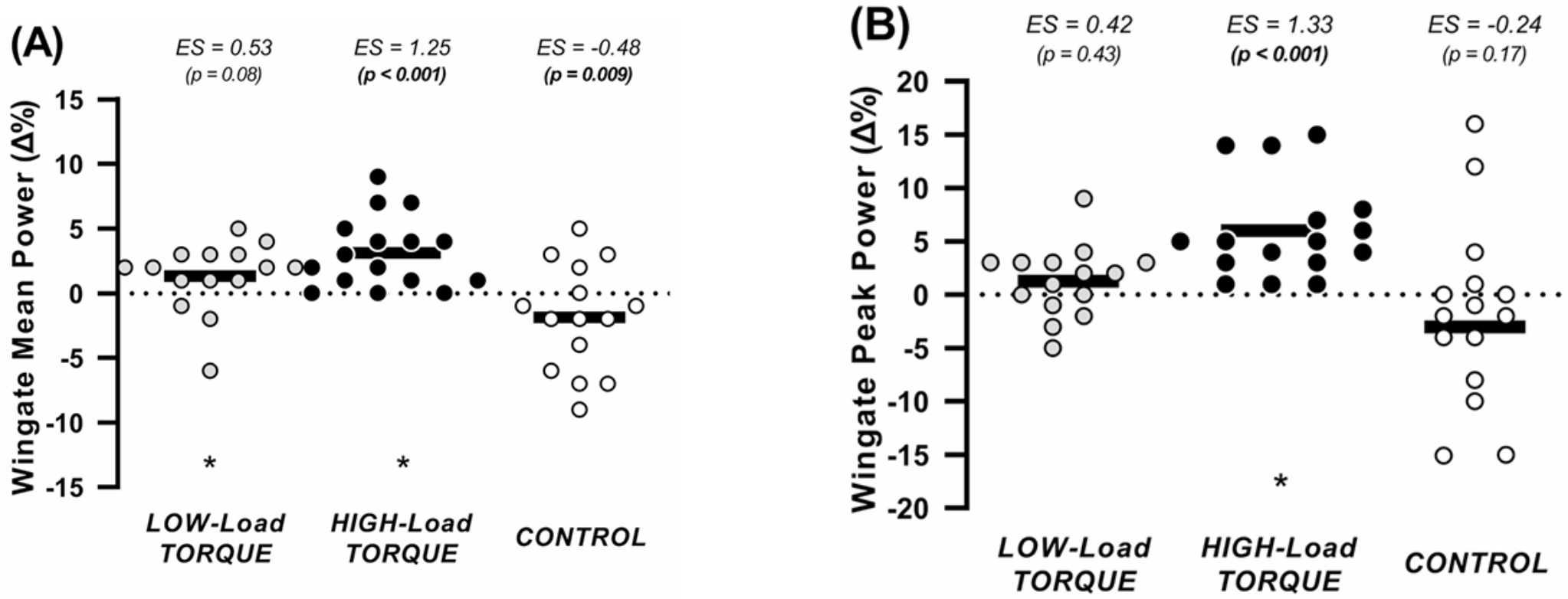




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(Barranco-Gil et al., 2025)

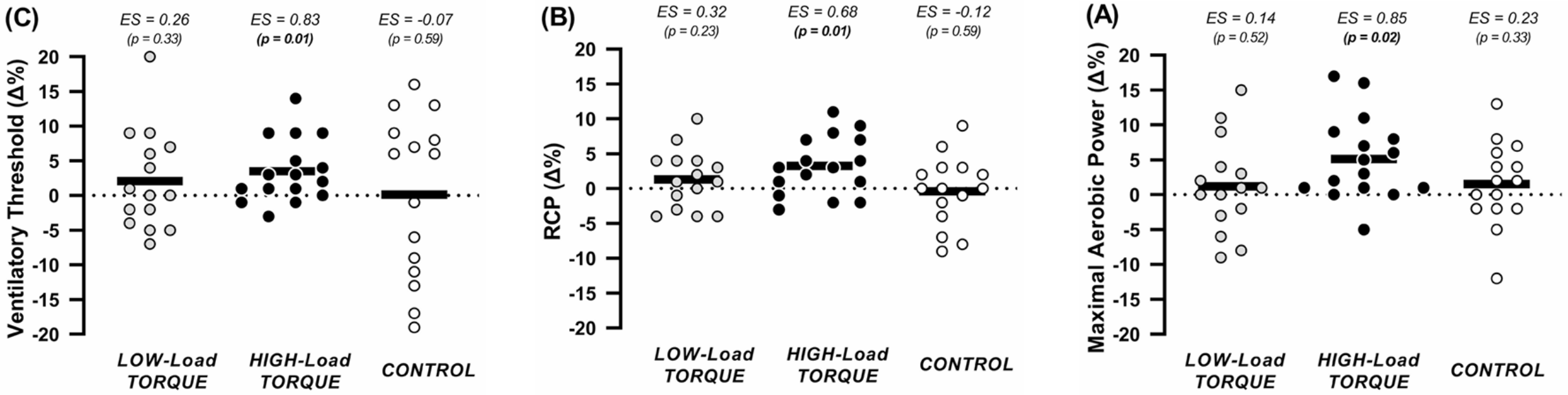
High vs low load torque



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(Pablos et al., 2026)

High vs low load torque



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(Pablos et al., 2026)

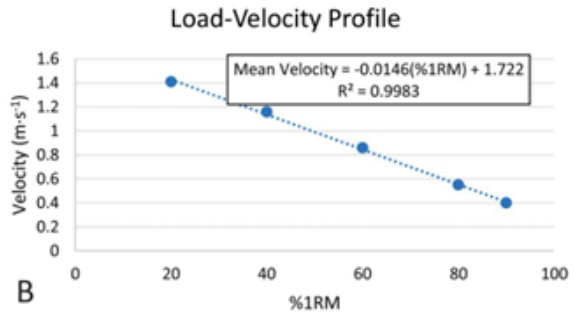
Testing 1RM

- Rechtstreeks 1RM testen
- Repitition based
- Velocity based

Author	1-RM prediction equation
Bryzcki (1993)	1-RM = 100 . rep wt / (102.78 – 2.78 . reps)
Epley (1985)	1-RM = (1 + .0333 . reps) . rep wt
Lander (1985)	1-RM = 100 . rep wt / (101.3 – 2.67123 . reps)
Lombardi (1989)	1-RM = rep wt . (reps)**.1
Mayhew et al. (1992)	1-RM = 100 . rep wt / (52.2+41,9 . exp (-.055 . reps))
O'Conner et al. (1989)	1-RM = rep wt (1 + .025 . reps)
Wathen (1994)	1-RM = 100 . rep wt / (48.8+53.8 . exp (-.075 . reps))

%1RM	Mean Velocity
20	1.41
40	1.16
60	0.86
80	0.55
90	0.40

A



B

Figure 2. (A) Mean velocity data attained from an athlete's L-V profile during the barbell back squat; (B) data, linear regression, and equation for this athlete's L-V profile.

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(Weakley et al., 2021)

Table 5. Challenges related to strength training, and perceived negative effects of strength training.

	Male n = 88	Female n = 59	Total n = 147	Sex differences	
Challenges to performing strength training	*n (%)*	*n (%)*	*n (%)*	*V*	*p-value*
Fatigue/soreness affecting the endurance training	46 (52.3)[a]	33 (55.9)[a,b]	79 (53.7)[a]	.036	n.s.
Limited time to perform strength training	20 (22.7)[b]	13 (22)[d]	34 (22.4)[b]	.008	n.s.
Traveling to races and training camps	49 (55.7)[a]	44 (74.6)[a]	93 (63.3)[a]	.235	<0.01*
Restarting strength training after stage races/racing periods	45 (51.1)[a]	29 (49.2)[b,c]	77 (50.3)[a]	.019	n.s.
Lack of knowledge on how to perform strength training	1 (1.1)[c]	2 (3.4)[e]	1 (2)[d]	.078	n.s.
Coach lacking knowledge on how to perform strength training	0 (0)[c]	2 (3.4)[e]	2 (1.4)[d]	.143	n.s.
Lack of interest/motivation to perform strength training	15 (17)[b]	1 (1.7)[e]	16 (10.9)[b,c]	.242	<0.01*
Lack of training equipment/facilities	18 (20.5)[b]	14 (23.7)[c,d]	32 (21.8)[b]	.039	n.s.
Other challenges	3 (3.4)[c]	2 (3.4)[e]	5 (3.4)[c,d]	.034	n.s.
Negative effects of strength training					
Impaired cycling performance	6 (6.8) [d]	1 (1.7) [c]	7 (4.8) [d]	.118	n.s.
Soreness impairing endurance training	64 (72.7) [a]	38 (64.4) [a]	102 (69.4) [a]	.088	n.s.
Increased risk of injury	19 (21.6) [c]	10 (16.9) [b,c]	29 (19.7) [c]	.057	n.s.
Increased muscle mass	42 (47.7) [b]	24 (40.7) [a,b]	66 (44.9) [b]	.069	n.s.
No negative effects	13 (14.8) [c,d]	11 (18.6) [b,c]	24 (16.3) [c,d]	.051	n.s.

Superscript letters indicate significant within-group differences using McNemar's test with Bonferroni correction. Variables that share at least one letter are not significantly different; those without a shared letter are significantly different. Superscripts are only comparable within each column (i.e., within-group). Abbreviations: *n* = total number of respondents selecting each, % = percentage of respondents, *V* = Cramer's *V*, n.s. = not significant.

(Vikestad et al., 2025)

Vermijden hypertrofie & vermoeidheid

Probleem

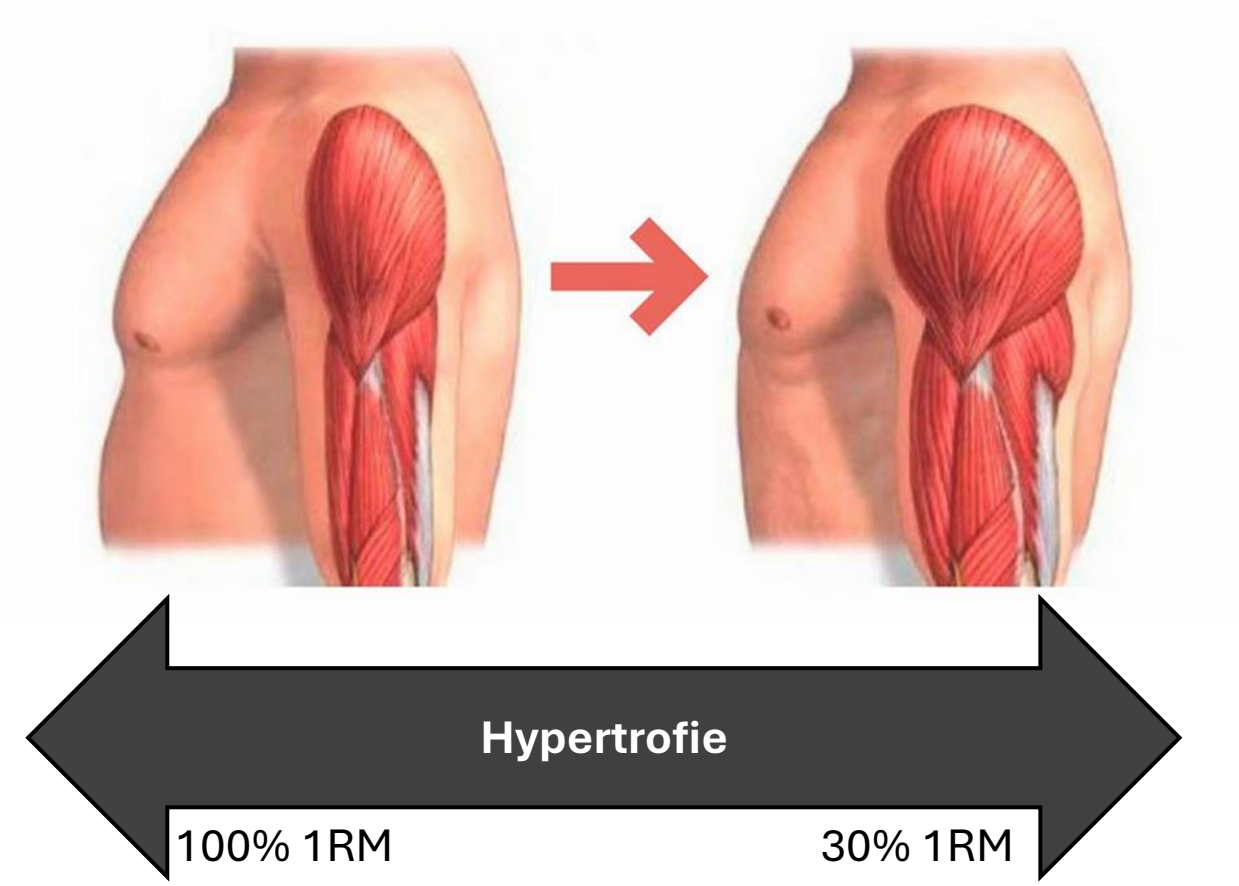
- Gewichtstoename
- Hoge vermoeidheid

Oorzaken

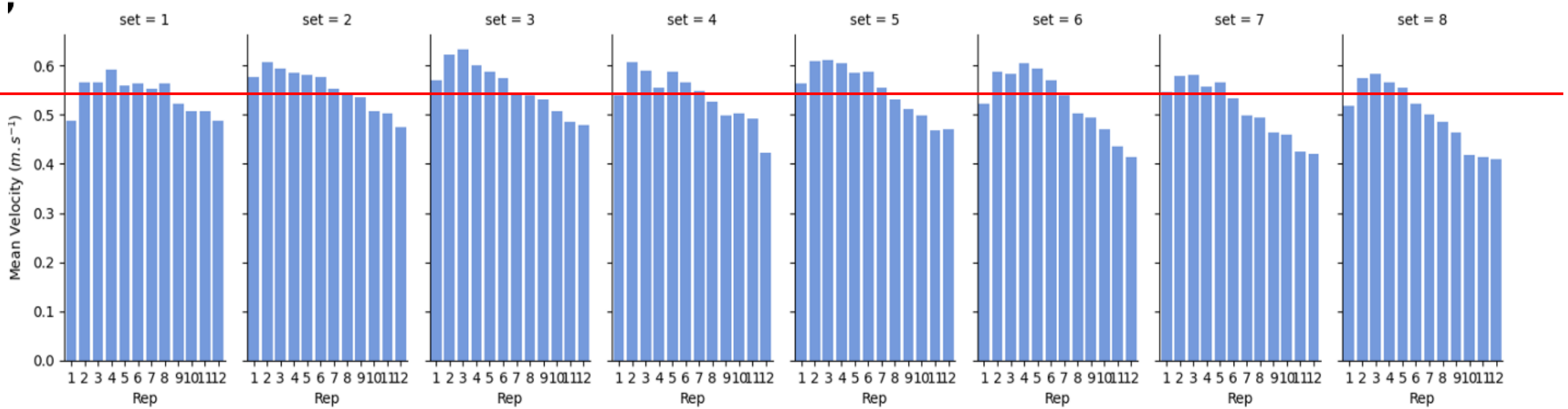
- Mechanische spanning
- Metabole prikkel
- Spierschade

Oplossing

↓ Volume

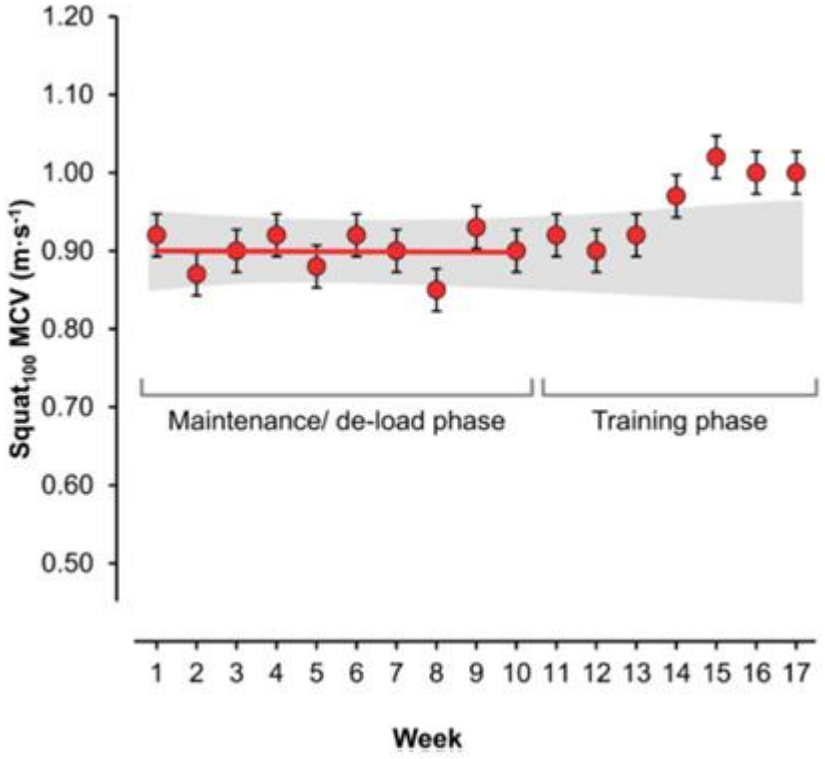


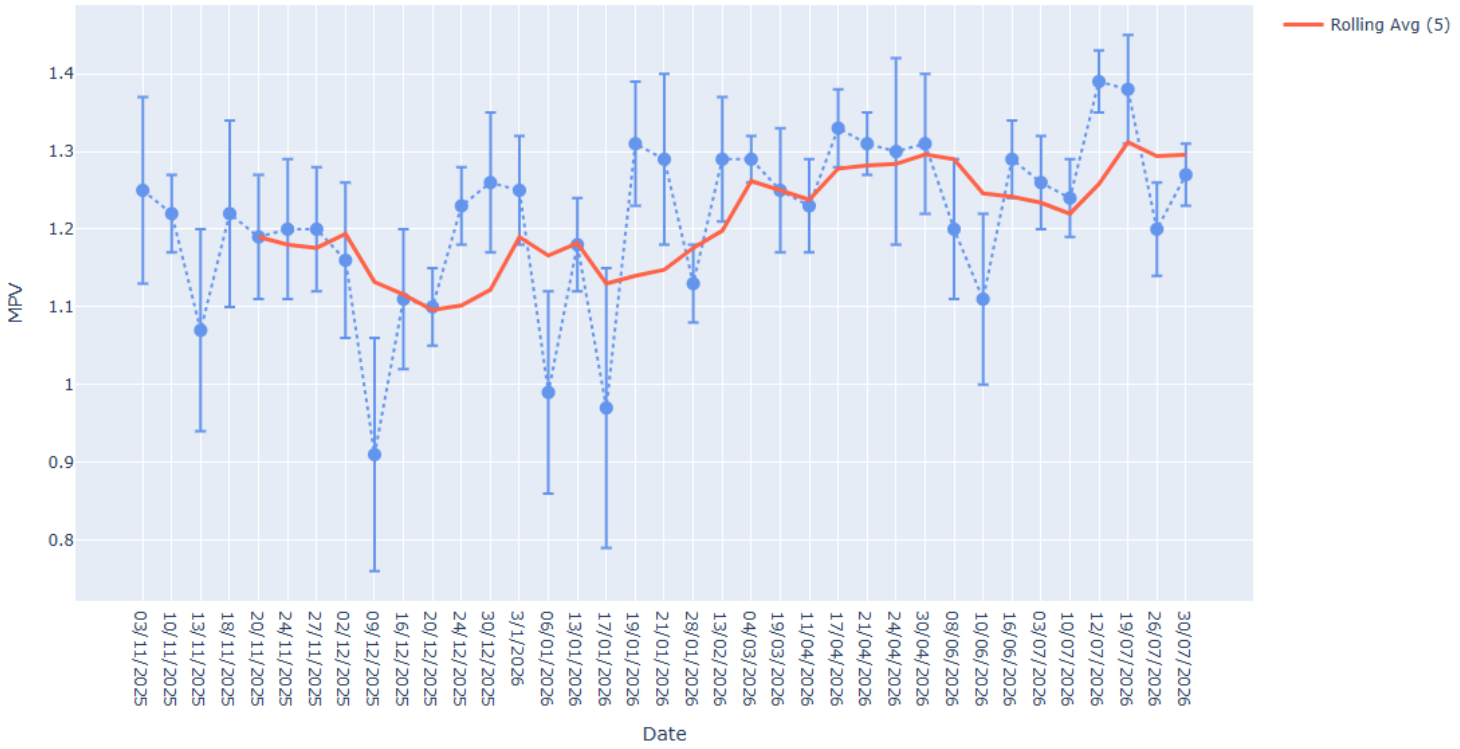
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Monitoring



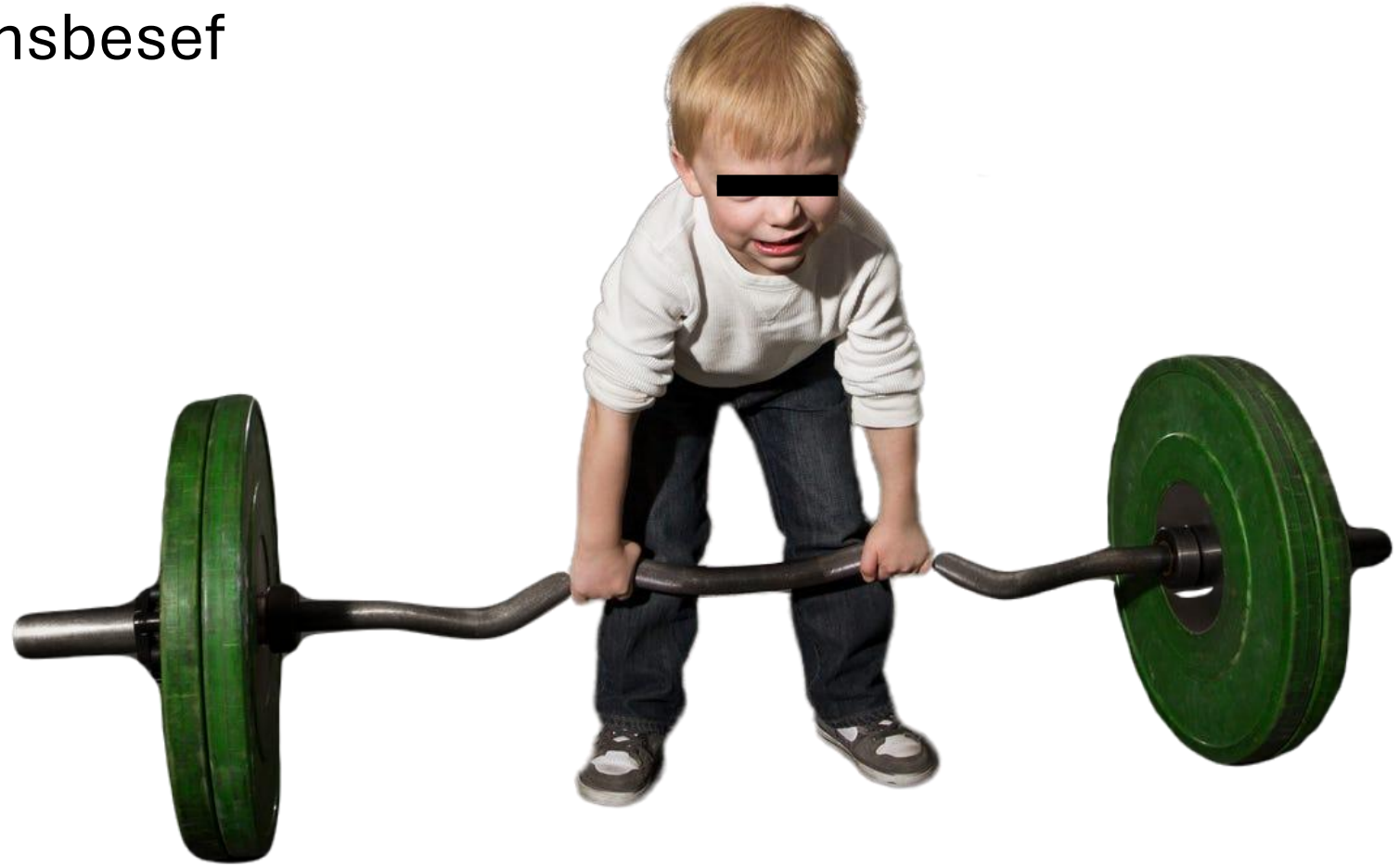


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(Weakley et al., 2021)

Krachttraining voor jeugdatleten

- Ontwikkeling → Lichaamsbesef
- Minder blessures
- Prestatie verbeteren



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Krachttraining voor jeugdatleten

< 12 jaar
Evenwicht
Motorische competentie
Leren bewegen
Mobiliteit

12 – 14 jaar
Posturale spieren
Kracht lichaamsgewicht
Leren techniek krachttraining

14 - 16 jaar
Posturale spieren
Eerste “echte” krachttraining
Leren techniek krachttraining
Algemene krachtoefeningen

> 16 jaar
Doorgedreven krachttraining
Sportspecifieke kracht

Techniek
- Squat (back, front, overhead)
- Deadlift
- Hip thrust
- (Clean, jerk, snatch)

Conditioneren
- Push ups
- Row (lichaamsgewicht)
- Lichaamsgewicht squat, lunge, step up, single leg squat, …
- Hoog aantal reps
- TL verhoge via volume

Techniek + klein gewicht
- Squat (back, front, overhead)
- Deadlift
- Hip thrust
- Lunge & step up
- (Clean, jerk, snatch)

Conditioneren
- Push ups, row (lichaamsgewicht)
- Licht gewicht (progressief): squat, lunge, step up, single leg squat, …
- TL verhoge via volume & gewicht

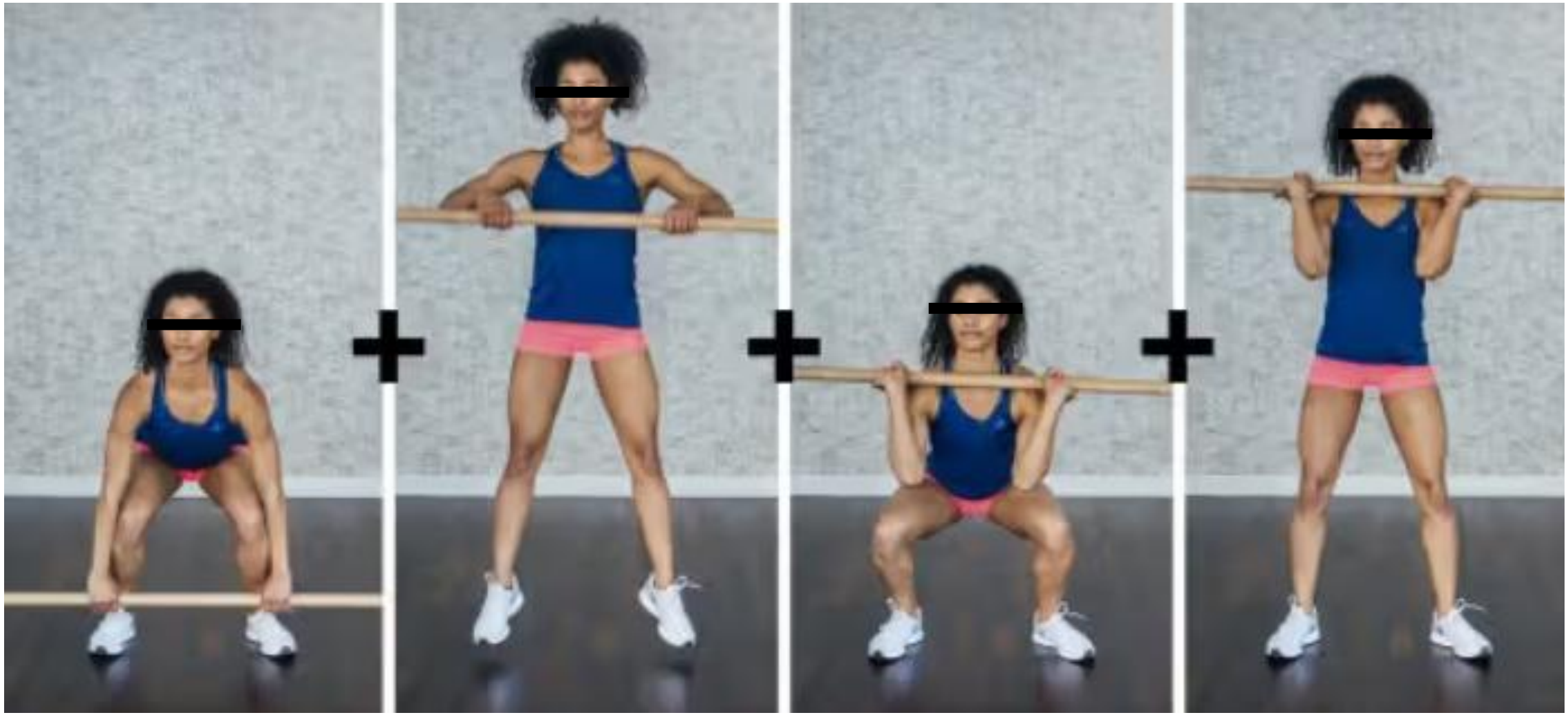
Doel
- Presteren/sterker worden
- Start volwaardig kracht programma

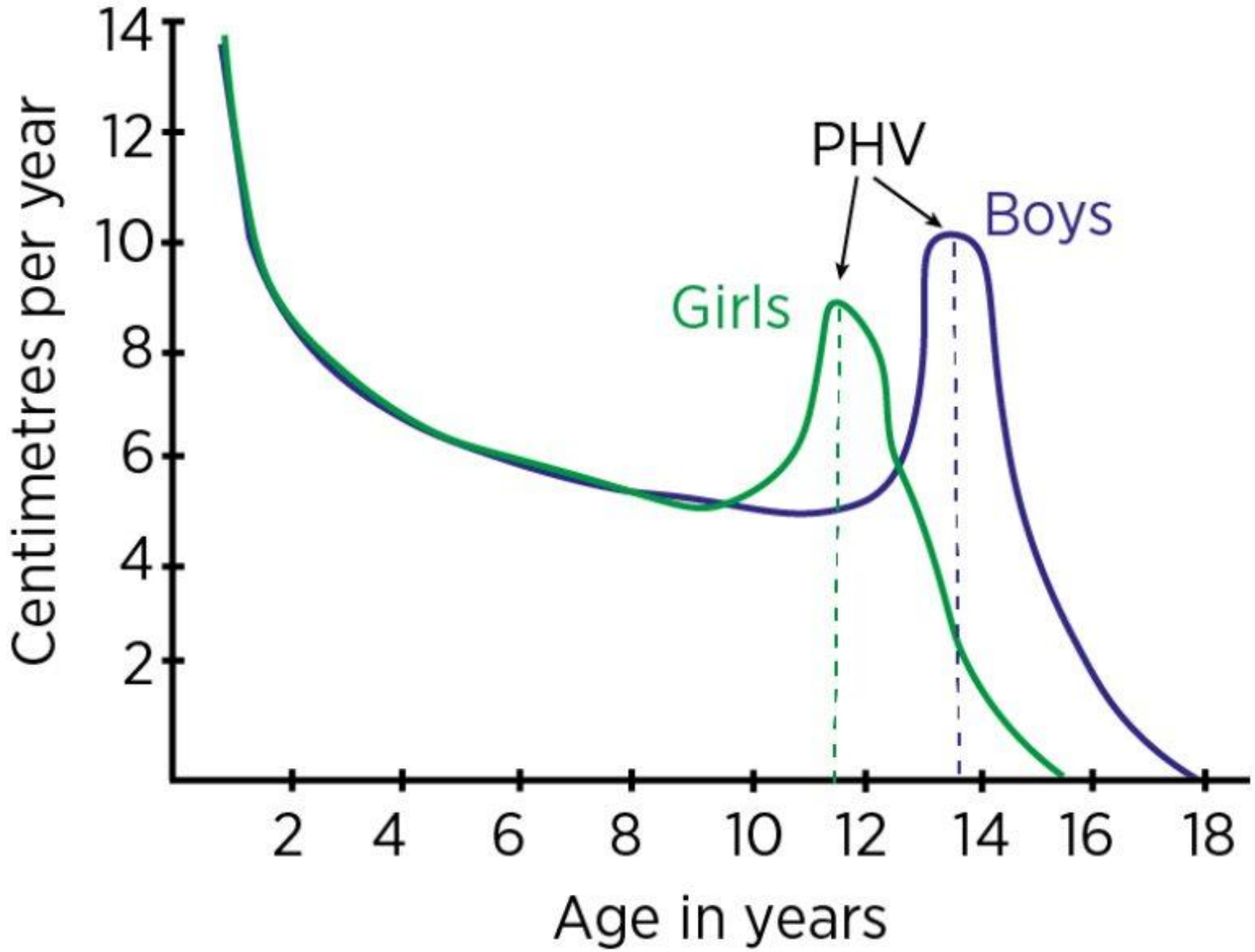
Tenzij nog geen ervaring!

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Centimetres per year
14
12
10
8
6
4
2
PHV
Boys
Girls
2
4
6
8
10
12
14
16
18
Age in years

2	3	4	5	6	7	8	Week 45
opt gemak bollen 01:30:00	**Kracht | Gym** 1) Landmine reverse lunge ---- sets: 3 ----- reps: 12 ---- gewicht: 7.5 2) Back squat --- sets:... 01:00:00	**Lange duur** 03:00:00	**rustig ochtendloopje als je kan** Rustig Duurtempo 00:30:00 **Kracht | Gym** 1) Landmine reverse lunge ---- sets: 3 ----- reps: 12 ---- gewicht: 7.5 2) Back squat --- sets:... 01:00:00	**race prep/verkenning** 01:01:20	**Wedstrijd** 01:00:00 190 bpm	**Lange duur** 03:00:00	00:00/12:01 0% 0/143 SS 0% 0/302 TSS® 0% 00:00/09:31 0% 0/143 SS 0% 0/302 TSS® 0% 00:00/00:30 0% 00:00/02:00 0%
14	15	16	17	18	19	20	Week 51
Kracht | Gym 2) Back squat --- sets: 3 ---- reps: 8 ----- gewicht: 25 7) Hip thrust--- sets: 2 ----- reps: 8... 01:00:00	**power sprints** 01:26:20	**Lange duur** 03:00:00	**Lange duur** 01:30:00	**Kracht | Gym** 2) Back squat --- sets: 3 ---- reps: 8 ----- gewicht: 25 7) Hip thrust--- sets: 2 ----- reps: 8... 01:00:00	**HIIIT** 01:45:00	**Lange duur** 03:00:00	00:00/12:41 0% 0/190 SS 0% 0/376 TSS® 0% 00:00/10:41 0% 0/190 SS 0% 0/376 TSS® 0% 00:00/02:00 0%

Vragen?

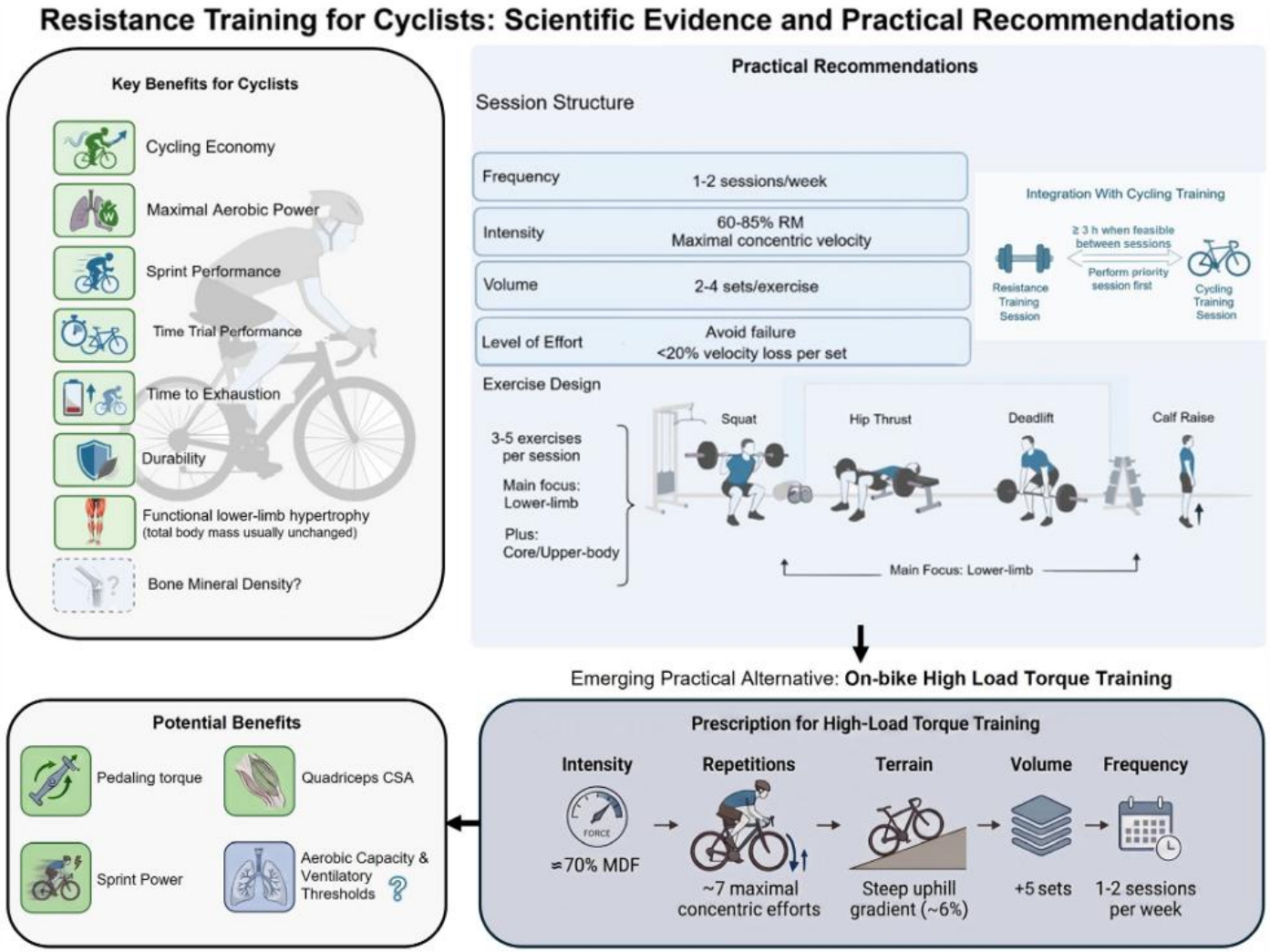


FIGURE 1 | Summary of the effects of resistance training and high-load torque training in cyclists, and evidence-based practical recommendations for their implementation. Of note, some of these are pragmatic recommendations derived by both cyclist-specific and broader evidence in the general population, and not all parameters have been directly established as optimal for cyclists.

de Pablos et al. 2026

Referenties

Vikestad, V., Lyngstad, I. K., & Dalen, T. (2025). Strength training among professional UCI road cyclists: Practices, challenges, and rationales. PLoS One, 20(7), e0328195. https://doi.org/10.1371/journal.pone.0328195

Joyner, M. J., & Coyle, E. F. (2008). Endurance exercise performance: the physiology of champions. J Physiol, 586(1), 35-44. https://doi.org/10.1113/jphysiol.2007.143834

García-Pallarés, J., & Izquierdo, M. (2011). Strategies to Optimize Concurrent Training of Strength and Aerobic Fitness for Rowing and Canoeing. Sports Medicine, 41(4), 329-343. https://doi.org/10.2165/11539690-000000000-00000

Beattie, K., Kenny, I. C., Lyons, M., & Carson, B. P. (2014). The effect of strength training on performance in endurance athletes. Sports Med, 44(6), 845-865. https://doi.org/10.1007/s40279-014-0157-y

Paavolainen, L., Häkkinen, K., Hämäläinen, I., Nummela, A., & Rusko, H. (1999). Explosive-strength training improves 5-km running time by improving running economy and muscle power. J Appl Physiol (1985), 86(5), 1527-1533. https://doi.org/10.1152/jappl.1999.86.5.1527

Aagaard, P., & Andersen, J. L. (2010). Effects of strength training on endurance capacity in top-level endurance athletes. Scand J Med Sci Sports, 20 Suppl 2, 39-47. https://doi.org/10.1111/j.1600-0838.2010.01197.x

Rønnestad, B. R., Hansen, E. A., & Raastad, T. (2010). Effect of heavy strength training on thigh muscle cross-sectional area, performance determinants, and performance in well-trained cyclists. Eur J Appl Physiol, 108(5), 965-975. https://doi.org/10.1007/s00421-009-1307-z

Rønnestad, B. R., Hansen, E. A., & Raastad, T. (2010). In-season strength maintenance training increases well-trained cyclists' performance. *European Journal of Applied Physiology, 110(6), 1269-1282. https://doi.org/10.1007/s00421-010-1622-4*

Schoenfeld, B. J., Grgic, J., Van Every, D. W., & Plotkin, D. L. (2021). Loading Recommendations for Muscle Strength, Hypertrophy, and Local Endurance: A Re-Examination of the Repetition Continuum. Sports (Basel), 9(2). https://doi.org/10.3390/sports9020032

Zatsiorsky, V. M., Kraemer, W. J., & Fry, A. C. (2020). Science and Practice of Strength Training. Human Kinetics. https://books.google.be/books?id=3v3FDwAAQBAJ

Bompa, T., & Buzzichelli, C. (2022). Periodization of Strength Training for Sports. https://doi.org/10.5040/9781718225428

Referenties

- *Rønnestad, B. R., Hansen, E. A., & Raastad, T. (2011). Strength training improves 5-min all-out performance following 185 min of cycling. Scand J Med Sci Sports, 21(2), 250-259. https://doi.org/10.1111/j.1600-0838.2009.01035.x*
- *Rønnestad, B. R., Hansen, J., Hollan, I., & Ellefsen, S. (2015). Strength training improves performance and pedaling characteristics in elite cyclists. Scand J Med Sci Sports, 25(1), e89-98. https://doi.org/10.1111/sms.12257*
- *Sunde, A., Støren, O., Bjerkaas, M., Larsen, M. H., Hoff, J., & Helgerud, J. (2010). Maximal strength training improves cycling economy in competitive cyclists. J Strength Cond Res, 24(8), 2157-2165. https://doi.org/10.1519/JSC.0b013e3181aeb16a*
- *Llanos-Lagos, C., Ramirez-Campillo, R., & Sáez de Villarreal, E. (2026). Heavy strength training effects on physiological determinants of endurance cyclist performance: a systematic review with meta-analysis. European Journal of Applied Physiology, 126(1), 193-222. https://doi.org/10.1007/s00421-025-05883-2*
- *Vikmoen, O., & Rønnestad, B. R. (2021). A Comparison of the Effect of Strength Training on Cycling Performance between Men and Women. J Funct Morphol Kinesiol, 6(1). https://doi.org/10.3390/jfmk6010029*
- *Weakley, J., Mann, B., Banyard, H., McLaren, S., Scott, T., & Garcia-Ramos, A. (2021). Velocity-Based Training: From Theory to Application. Strength & Conditioning Journal, 43(2), 31-49. https://doi.org/10.1519/ssc.0000000000000560*
- *Barranco-Gil, D., Hernández-Belmonte, A., Rodríguez-Rielves, V., Iriberri, J., Martínez-Cava, A., Buendía-Romero, Á., Alejo, L. B., Franco-Lopez, F., Sanchez-Redondo, I. R., R, D. E. P., Lucia, A., Valenzuela, P. L., & Pallares, J. G. (2025). Off- and On-Bike Resistance Training in Cyclists: A Randomized Controlled Trial. Med Sci Sports Exerc, 57(2), 296-304. https://doi.org/10.1249/mss.0000000000003556*
- *Pablos, R., Valenzuela, P. L., Martínez Cava, A., B. Alejo, L., Revuelta, C., Sánchez-Redondo, I., Barranco-Gil, D., & Pallarés, J. (2026). The Role of Pedaling Intensity During Torque Training in Cyclists: A Three-Arm Randomized Controlled Trial. Scandinavian Journal of Medicine & Science in Sports, 36. https://doi.org/10.1111/sms.70294*
- *de Pablos, R., Sánchez-Redondo, I. R., Alejo, L. B., Pallarés, J. G., Barranco-Gil, D., & Valenzuela, P. L. (2026). Resistance Training for Cyclists: Scientific Evidence and Practical Recommendations. Scandinavian Journal of Medicine & Science in Sports, 36(10), e70377. https://doi.org/https://doi.org/10.1111/sms.70377*